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Safe formation scaling and motion planning for heterogeneous UAV–UGV teams in cluttered environments

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Heterogeneous UAV–UGV formations offer significant potential for autonomous missions; however, achieving stable three-dimensional (3-D) formation control with collision-free navigation in dense obstacle environments remains challenging due to heterogeneous dynamics, unknown external disturbances, and strict real-time computational constraints. This paper proposes a distributed rigid graph-based adaptive safe artificial potential field (RG–ASAPF) framework that tightly integrates formation control and motion planning for heterogeneous multi-UAV–UGV systems. The proposed framework addresses three key challenges: (i) maintaining continuous formation integrity during obstacle avoidance, thereby avoiding the conventional disassembly-recovery process; (ii) enabling vertical coordination between UAV and UGV layers with prescribed inter-layer spacing; and (iii) supporting dynamic formation reconfiguration through time-varying scaling to safely navigate narrow passages. A backstepping-based adaptive formation controller is developed using rigid graph theory to enforce distance consensus under Euler-Lagrange dynamics, explicitly accounting for underactuated UAVs and nonholonomic UGVs. An adaptive safe artificial potential field with Widrow–Hoff-based online repulsion tuning generates smooth, collision-free trajectories while respecting formation footprint constraints. Rigorous Lyapunov analysis establishes exponential convergence of formation errors and global uniform ultimate boundedness (GUUB) of tracking errors in the presence of bounded disturbances. Furthermore, a safety-certified fallback mechanism with corridor-aware rerouting enhances robustness when clearance constraints are violated. Extensive simulations involving sparse and dense obstacle environments, formation scaling maneuvers, dynamic obstacles, and formation maneuvers featuring more agents demonstrate that the proposed framework outperforms model predictive control, particle swarm optimization, and conventional artificial potential field methods in terms of computational efficiency, trajectory smoothness, formation stability, and energy consumption. These results highlight the effectiveness of the proposed approach for heterogeneous multi-domain robotic applications such as disaster response, surveillance, and cooperative logistics.

Keywords Heterogeneous UAV–UGV cooperation, Time-varying formation control, Motion planning, Graph theory, Adaptive safe artificial potential field

Heterogeneous cooperative robotic systems composed of unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) have attracted increasing attention in recent years due to their complementary sensing, mobility, and payload capabilities[1–5]. Such UAV–UGV teams, as illustrated in Fig. 1, have demonstrated strong potential in surveillance, search and rescue, infrastructure inspection, and disaster response, where complex environments, limited accessibility, and safety constraints demand coordinated multi-agent behaviors rather than single-robot operation[6–9].

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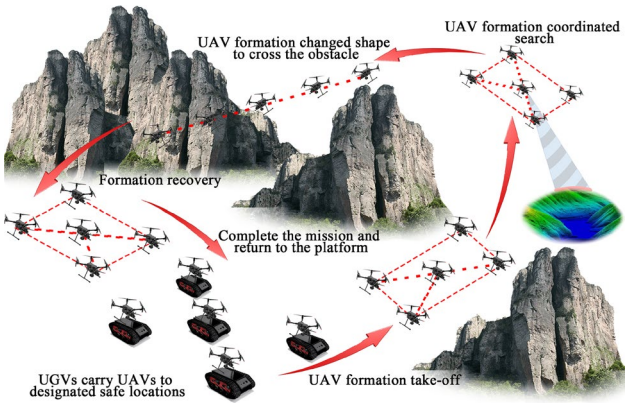


Fig. 1. The schematic diagram of cooperative UAV and UGV systems.

Despite this potential, achieving stable three-dimensional (3-D) formation motion planning with guaranteed collision avoidance for heterogeneous UAV–UGV systems in cluttered environments remains a challenging problem[10–13]. First, UAVs and UGVs exhibit fundamentally different kinematic and dynamic characteristics: UAVs are typically underactuated and operate in 3-D space, whereas UGVs are subject to nonholonomic constraints and terrain-induced disturbances[14]. Designing a unified control framework that preserves formation geometry while respecting these heterogeneous constraints is nontrivial. Second, realistic operating environments often contain dense static and dynamic obstacles, external disturbances (e.g., wind, uneven terrain), and limited sensing ranges, which significantly complicate real-time motion planning and stability guarantees[15]. Third, practical missions frequently require formation reconfiguration, such as continuous scaling or reshaping, to safely traverse narrow passages or obstacle clusters–yet most existing methods treat formation changes as discrete events without rigorous stability analysis.

Existing formation control approaches, including leader–follower schemes, virtual structure methods, and graph-theoretic strategies, provide effective coordination mechanisms under ideal conditions[16,17]. However, many of these approaches decouple formation maintenance from obstacle avoidance, leading to inevitable formation disassembly and reassembly during navigation in cluttered environments. Artificial potential field (APF)–based methods are computationally attractive for real-time obstacle avoidance, but they are well known to suffer from local minima, oscillations near obstacles, and sensitivity to parameter tuning, especially when applied to formation–level motion planning rather than single–agent navigation[18]. More advanced planning techniques such as model predictive control (MPC), particle swarm optimization (PSO), and reinforcement learning can improve path optimality, but often incur prohibitive computational costs or lack formal stability guarantees for large-scale heterogeneous formations[19].

A careful review of the literature reveals several critical gaps.

- Most existing UAV–UGV formation methods do not preserve continuous formation integrity during obstacle avoidance, resulting in fragmented motion and degraded coordination[20,21].
- Vertical coordination between UAV and UGV layers is often neglected, limiting truly 3-D formation consistency[22].
- Formation scaling or reshaping is rarely addressed within a rigidity-preserving framework with explicit stability guarantees[23].
- Safety mechanisms for extreme clutter or narrow passages, such as fallback or replanning strategies, are seldom formalized.

To address these challenges, this paper proposes a rigid-graph-based adaptive safe artificial potential field (RG-ASAPF) framework for heterogeneous UAV–UGV formation motion planning. By integrating rigid graph theory, backstepping-based adaptive formation control, and an adaptive safe artificial potential field, the proposed framework enables continuous formation maintenance, time-varying scaling, and collision-free navigation in dense 3-D environments. A bounded Widrow–Hoff adaptation law is employed to tune repulsive potentials online, ensuring safety while avoiding excessive conservatism. Furthermore, a safety-certified pause-and-reroute fallback mechanism is incorporated to handle extreme clutter scenarios where formation footprint constraints are temporarily violated.

The main contributions of this work are summarized as follows:

- A unified rigid-graph-based formation control framework for heterogeneous UAV–UGV systems that guarantees continuous 3-D formation integrity under time-varying scaling maneuvers.
- An adaptive safe artificial potential field (ASAPF) with bounded online tuning, explicitly accounting for formation footprint and obstacle proximity.
- A coupled planner–controller stability analysis establishing exponential convergence of formation errors and global uniform ultimate boundedness (GUUB) under bounded disturbances.

- Extensive quantitative simulations demonstrating superior computational efficiency, trajectory smoothness, formation stability, and energy performance compared with conventional APF-, improved APF-, PSO-, MPC-, and NMPC-based methods.

The remainder of this paper is organized as follows. Section reviews the related literature on formation control and motion planning for heterogeneous Multi-agent systems (MASs). Section formulates the system model and presents tha essential preliminaries on graph theory and Euler-Lagrange dynamics. Section introduces the proposed formation control law with adaptive parameter estimation grounded in rigid graph theory. Section presents the adaptive safe artificial potential field motion-planning algorithm. Section provides simulation results and a comparative performance analysis under diverse scenarios. Finally, Section concludes the paper and discusses directions for future research.

Related work

The cooperative control of heterogeneous MASs, particularly UAV–UGV formations, has attracted significant research attention over the past two decades. This section provides a critical review of existing formation control strategies and motion-planning algorithms, highlighting their technical contributions, inherent limitations, and the research gaps that motivate the present work.

Formation control of heterogeneous UAV–UGV systems

Formation control of heterogeneous MASs has been extensively investigated using the leader–follower strategies[24,25], the virtual structure methods[26], the behavior-based method[27], the graph-theoretic approaches[28], the potential field method[29], and the reinforcement learning-based solutions[30]. Graph-based formation control, in particular, provides a rigorous mathematical framework for maintaining inter-agent distances and has been successfully applied to homogeneous and heterogeneous multi-agent systems. Rigid graph theory further enables distance-based formation maintenance with provable stability guarantees and scalability. Table 1 summarizes these strategies, illustrating that no single approach simultaneously satisfies all desirable properties: formation acquisition, trajectory tracking, reconfiguration, obstacle avoidance, stability, real-time performance, and computational efficiency.

A growing body of literature has investigated formation control specifically for heterogeneous UAV–UGV systems[31–35]. Rabelo et al.[31] presented a virtual structure-based scheme for UAV landing on static or mobile UGVs, addressing tight ai–ground coordination requirements. Zhao et al.[32] developed a data-driven robust formation controller leveraging local information and reinforcement learning for optimal group control, though explicit obstacle avoidance was not considered. Shi et al.[33] proposed a hierarchical learning framework for formation tracking but did not account for environmental obstacles. Gong et al.[34] introduced a distributed adaptive formation control method with prescribed performance for actuator failures, yet formation reconfiguration capabilities were not addressed. Babazadeh and Selmic[35] developed a distance-based distributed strategy for heterogeneous nonlinear agents, demonstrating scalability. However, despite these advances, most existing approaches focus primarily on formation stabilization or tracking under nominal conditions and assume obstacle-free environments or simplified avoidance strategies. As a result, formation integrity is often compromised when obstacle avoidance is required, leading to formation disassembly and subsequent reassembly, which degrades coordination efficiency and safety.

Motion planning and trajectory optimization for multi-agent formations

Trajectory planning is critical for enabling MAS formations to navigate complex environments while avoiding collisions and maintaining formation geometry. Planning algorithms are broadly classified into *global* (offline) and *local* (online) methods. Global approaches include traditional algorithms such as Dijkstra, Rapid-exploration Random Tree (RRT), Probabilistic Road Map (PRM), and Dubins curves, as well as intelligent algorithms such as PSO, A*, Salp Swarm Algorithm (SSA), Artificial Bee Colony (ABC), Neural Networks (NN), and Reinforcement Learning (RL)[36–40]. While these methods can achieve optimal or near-optimal paths, many demand substantial computational resources, particularly in high-dimensional 3-D environments, rendering them unsuitable for real-time deployment[41]. Local planning methods include APF, MPC, Mathematical Optimization Algorithms (MOA), Dynamic Window Approach (DWA), and Bug algorithms. Among these, APF methods are widely recognized for their computational efficiency and simplicity in real-time obstacle avoidance. Classical APF approaches generate repulsive forces based on obstacle proximity and attractive forces toward the goal, making them attractive for single-agent navigation and low-complexity systems[42].

Description	Leader-follower	Virtual structure	Behavior-based	Graph based	Potential field-based	Reinforcement learning
Formation acquisition	✗	✗	✓	✓	✓	✓
Formation trajectory tracking	✓	✓	✓	✓	✓	✓
Formation reconfiguration	✗	✗	✗	✓	✓	✓
Obstacle avoidance	✓	✗	✓	✗	✓	✓
Formation stability	Medium	High	Low	High	Medium	Medium
Real-time performance	High	Low	High	Medium	High	Low
Computational complexity	Low	High	High	Medium	Low	High

Table 1. Comparison between formation control strategies. Able (✓), Unable (✗)

Despite their popularity, APF methods suffer from several well-known drawbacks, including local minima, oscillatory behavior near obstacles, and sensitivity to parameter tuning. These issues become more pronounced when APF techniques are extended from single-agent navigation to formation-level planning, where formation footprint, inter-agent safety, and rigidity constraints must be simultaneously considered.

For heterogeneous UAV–UGV systems, several specialized motion-planning approaches have been proposed. Niu et al.[43] developed a vision-based UAV-assisted system for collision-aware obstacle avoidance trajectory generation for UGVs. However, this work merely studied the UAV-assisted UGV in planning maneuvering paths and did not involve formation control issues. Ding et al.[44] designed a memetic algorithm combining genetic search operators and gradient-based repair mechanisms for UAV path optimization, but this study did not take into account the interaction between UAVs and UGVs. Wu et al.[45] proposed a hybrid distribution estimation and genetic algorithm for UAV–UGV cooperative path planning in urban environments. This algorithm is more focused on solving multi-point optimization problems and is not very suitable for the collaborative path planning problem of heterogeneous systems. Bacheti et al.[46] introduced an obstacle avoidance algorithm for path-following tasks in UAV–UGV formations. Although this study has taken into account both single-obstacle and multi-obstacle environmental conditions, the number of formation agents is too small, which greatly reduces the complexity of the research. Adderson and Pan[47] presented a continuously varying formation framework with novel potential field avoidance, though formation configuration could not be consistently maintained throughout obstacle avoidance. Although this study has considered the motion planning problem of heterogeneous formation systems in single-obstacle and multi-obstacle environments, it has not taken into account the influence of dynamic obstacles on formation structure and planned paths. Liu et al.[48] combined consensus-based formation control with jump point search for global planning and local APF/DWA methods for real-time obstacle avoidance. However, this study is based on traditional APF to solve the formation obstacle avoidance problem in a simple obstacle environment, without considering multiple obstacles and external environmental disturbances. Wu et al.[49] designed a hybrid genetic-simulated annealing algorithm for task allocation and multistep PSO strategy for collision-free path generation, though formation integrity was not preserved during obstacle avoidance. However, this study only considered heterogeneous UAV systems and did not take into account UGVs or other heterogeneous agents. Table 2 compares these trajectory planning algorithms across multiple criteria, revealing that while intelligent algorithms offer strong optimization and high-quality paths, they incur substantial computational costs. Conversely, APF-based methods provide real-time performance but suffer from limited path optimality and potential local minima.

Beyond theoretical frameworks, numerous researchers have conducted empirical investigations to verify the practical viability of proposed algorithms. Yan et al.[50] designed a secure time-varying formation control strategy addressing denial-of-service attacks, non-cooperative dynamic barriers, and input saturation, which was validated through hardware experiments. However, in this work, each agent avoided obstacles independently, and the integrity of the formation was disrupted during the obstacle avoidance process. Xiong et al.[51] investigated the issue of fully distributed optimal tracking control of UAV and UGV formation under Byzantine attacks, where an optimal control method under the reinforcement learning (RL) algorithm is investigated to against the aggressive Byzantine attacks. Finally, simulation and experimental results are carried out to demonstrate the efficacy of the proposed approach. However, the problem of formation obstacle avoidance were not been studied. Li et al.[52] proposed a hierarchical cross-domain formation strategy for heterogeneous UAV–UGVs to realize time-varying formation trajectory tracking with uncertainties, where both Gazebo simulation and physical experiment verify the effectiveness of the theoretical results. However, this work also did not take into account the issue of formation obstacle avoidance. Chen et al.[53] developed a heterogeneous leader-follower framework with angle-specific constraints leveraging relative bearing measurements,and validated the proposed method via physical experiments. However, this work did not take into account the impact of external disturbances

Algorithm	Static/Dynamic obstacle	Scenario	Coverage	Computational complexity	Path optimization	Path smoothness
Djikstra	Static	2-D	Off-line	High	Strong	Weak
RRT	Static	2-D/3-D	Off-line	High	Weak	Weak
PRM	Static	2-D/3-D	On-line	High	Weak	Weak
Dubins curve	Static	2-D	On-line	High	Strong	Strong
PSO	Static	2-D	On-line	High	Moderate	Strong
A*	Static	2-D/3-D	On-line	High	Strong	Weak
SSA	Static/Dynamic	2-D	Off-line	High	Weak	Strong
ABC	Static	2-D	Off-line	High	Weak	Strong
NN	Static	2-D/3-D	Off-line	High	Moderate	Moderate
RL	Static/Dynamic	2-D/3-D	On-line	High	Strong	Moderate
APF	Static	2-D/3-D	On-line	Low	Weak	Strong
MPC	Static/Dynamic	2-D/3-D	On-line	High	Strong	Strong
MOA	Static	2-D	Off-line	High	Weak	Weak
DWA	Static/Dynamic	2-D	On-line	Low	Weak	Strong
Bug Algorithm	Static	2-D	On-line	Low	Weak	Weak

Table 2. Comparison between some trajectory planning algorithms.

on the formation. Martínez-Rozas et al.[54] addressed the problem of trajectory planning in a marsupial robotic system consisting of an UAV linked to an UGV through a non-taut tether with controllable length, the method was based on optimal Rapidly-exploring Random Trees (RRT*) with novel sampling and steering techniques to speed-up the computation. However, this work did not take into account the influence of dynamic obstacles on the algorithm. Despite these contributions, experimental validations remain relatively limited, and most studies do not explicitly address the integrated design of formation control and motion planning under realistic external disturbances and dense obstacle environments.

Summary of gaps and motivation for this work

Despite significant achievements in heterogeneous UAV–UGV formation control and motion planning, critical gaps still persist. *First*, advanced planning techniques such as MPC, PSO, and learning-based methods have been applied to multi-agent formation navigation to improve path optimality and adaptability. MPC-based approaches can explicitly handle constraints and predict future system behavior, while PSO and reinforcement learning methods can explore complex solution spaces. However, these methods typically incur high computational costs, making real-time implementation challenging for large-scale heterogeneous formations. In addition, many learning-based approaches lack formal stability guarantees and require extensive training data, which limits their applicability in safety-critical missions. Furthermore, formation integrity and rigidity are often not explicitly enforced during optimization, leading to suboptimal coordination performance. *Second*, although the principle of the APF algorithm is simple and popularity, APF methods suffer from several well-known drawbacks, including local minima, oscillatory behavior near obstacles, and sensitivity to parameter tuning. These issues become more pronounced when APF techniques are extended from single-agent navigation to formation-level planning, where formation footprint, inter-agent safety, and rigidity constraints must be simultaneously considered. *Third*, formation reconfiguration, including scaling and reshaping, is essential for navigating narrow passages and dense obstacle environments. Existing approaches typically implement reconfiguration through discrete switching strategies, predefined formation templates, or optimization-based planners. While these methods can improve navigability, they often introduce abrupt geometry changes that may destabilize the formation or violate actuator and safety constraints.*Forth*, vertical UAV–UGV coordination in fully 3-D environments is insufficiently addressed.*Fifth*, safety mechanisms for extreme clutter and narrow passages are rarely formalized. Motivated by these gaps, this paper develops a rigid-graph-based adaptive safe artificial potential field (RG-ASAPF) framework that unifies formation control, formation-aware motion planning, continuous scaling, and safety-certified fallback mechanisms within a single stability-guaranteed architecture. This distinguishes the proposed approach from existing methods and advances the state of the art in heterogeneous multi-UAV–UGV formation navigation.

Preliminaries

In the present section, some preliminaries are given. The notations are introduced first, and the cooperative control models of heterogeneous multi-UAV–UGV formation systems are then discussed. Subsequently, basic concepts of graph theory and APF are are presented. Finally, the problem formulation and control objectives are stated.

Nomenclature

The definitions of key symbols and nomenclature for the heterogeneous multi-UAV–UGV formation system are presented in Table 3.

Model of UAV and UGV

Consider a heterogeneous multi-UAV–UGV system consisting of *N* agents navigating through a dense obstacle environment to execute specific missions. Assume that there exist *M* UAVs and $N-M$ UGVs, among which a UGV that can comprehensively process environment information and plan motion paths is defined as the leader UGV, and other agents that execute tasks are defined as followers. In this article, the design of the formation controller is based on the distance between multiple agents, therefore, we assume that the formation structure only depends on the positions of multiple agents. Accordingly, the formation control model is simplified to a position-based structure. This paper studies the motion of various agents at both the formation control and motion planning levels. Thus, we consider the kinematic and dynamical models of underactuated UAVs and nonholonomic constraints of UGVs, which are selected from[11,55]. As illustrated in Fig. 2, the reference frame $\{E\}=\{X_0,Y_0,Z_0\}$ denotes the earth-fixed coordinate frame, while $\{B\}=\{X_i,Y_i,Z_i\}$ represents the body-fixed coordinate frame of the *i*th agent, attached to the agent's center of mass.

1) UAV Model: The underactuated UAV is utilized in this heterogeneous system, and the kinematic of *i*th UAV can be described by[32].

$$\begin{cases} \dot{x}_{ci} = v_i c\beta_i c\alpha_i \\ \dot{y}_{ci} = v_i c\beta_i s\alpha_i \\ \dot{z}_{ci} = v_i s\beta_i \\ \dot{\alpha}_i = w_{\alpha i} \\ \dot{\beta} = w_{\beta i}, \end{cases} \tag{1}$$

in which, $c\beta_i \triangleq \cos\beta_i$, $c\alpha_i \triangleq \cos\alpha_i$, $s\beta_i \triangleq \sin\beta_i$, $s\alpha_i \triangleq \sin\alpha_i$. $p_{ci}=[x_{ci},y_{ci},z_{ci}]^T$ denotes the position coordinate of the center of mass for the *i*th UAV, α_i and β_i stand for the heading angles of body-fixed coordinate frame $\{B\}$ relative to earth-fixed coordinate frame $\{E\}$, v_i represent the UAV linear velocity, and $w_{\alpha i}$ and $w_{\beta i}$ denote the angular velocities. Correspondingly, the dynamic model of the *i*th UAV can be described as

Symbol	Definition	Symbol	Definition
$\mathbb{R}^N$	N-dimensional Euclidean space	d_{safe}	Safe distance
$\mathbb{R}^{m\times n}$	$m \times n$ real matrix space	e_{ij}	Distance error
I_N	N-dimensional identity matrix	σ_i	Alternative error variable
1_N	1-column vector	W_{ij}	Lyapunov function
$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$	A undirected graph	$\hat{\phi}_i$	Parameter estimation of ϕ_i
$\mathcal{V} = \{1, 2, \ldots, N\}$	The set of agents	$\tilde{\phi}_i$	Parameter estimation error
$\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$	The edge set	v_f	Fictitious speed input
$\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N\times N}$	The adjacency matrix	v_d	Intended navigation speed of agent
$\mathcal{F}$	Formation framework	s	Introduced velocity variable
$\mathcal{N}_i$	Neighbor set of the agent i	W_d	Updated Lyapunov function
N	Number of agents	U_{att}	Attractive potential field function
M_i	Inertia matrix	F_{att}	Attractive force
C_i	Dentripetal/Coriolis terms matrix	U_{rep}	Repulsive potential field function
D_i	Damping matrix	F_{rep}	Repulsive force
p_i	Position of agent i	k_{att}	Attraction coefficient
α_i, β_i	Heading angle of UAV i	k_{rep}	Repulsive coefficient
$J(\cdot)$	Conversion matrix	ρ_0	Influence radius of the obstacle
θ_i	Heading angle of UGV i	ρ_g	Influence radius of the goal
v_i	Linear velocity of agent i	K	Number of obstacles
τ_i	Control input of agent i	p_{obs}	Position of obstacle
Y_i	Regression matrix	p_{goal}	Position of goal
ϕ_i	Model parameter	k_a, k_v, k_m	Parameters of formation controller
$R(p)$	Rigidity matrix	$diag\{\cdot\}$	Diagonal matrix
$d_{ij}(t)$	Desired distance between two agent i and j	A^T	transposition of the matrix A
p_v	Position of planned waypoint	$\otimes$	Kronecker product
$\tilde{p}_{ij}$	Distance between two agent i and j	$\|\cdot\|$	2-norm of a vector

Table 3. Definitions of key symbols and nomenclature for the heterogeneous multi-UAV–UGV formation system.

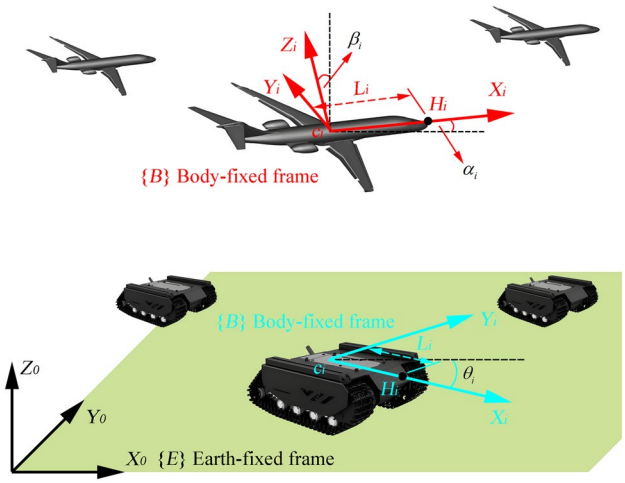


Fig. 2. Schematic diagram of coordinate transformation of UAV and UGV.

$$\bar{M}_i\dot{\eta}_i + \bar{D}_i\eta_i = \tau_i, \tag{2}$$

where $\eta_i = [v_i, w_{\alpha i}, w_{\beta i}]^T$ denotes the vector of linear speed and angular velocity of the ith UAV, $\bar{M}_i = diag\{m_i, I_{1i}, I_{2i}\}$ with m_i represents the mass, I_{1i} and I_{2i} denote the inertial moments of the ith UAV, respectively. $\bar{D}_i \in \mathbb{R}^{3\times3}$ denotes the constant damping matrix, while $\tau_i \in \mathbb{R}^3$ stands for the force or torque level control input. Inspired by[11,55], the following coordinate transformation introduced to solve the underactuated problem, this transformation can linearize the dynamics. As shown in Fig. 2, the "hand" point H_i, with position $p_i = [x_i, y_i, z_i]^T \in \mathbb{R}^3$, which is at a fixed offset L_i from the center of mass p_{ci}, is defined as $p_i = p_{ci} + L_i[c\beta_i c\alpha_i, c\beta_i s\alpha_i, s\beta_i]^T$, and taking the time derivative of p_i, one have

$$\dot{p}_i = \begin{bmatrix} c\beta_i c\alpha_i & -L_i c\beta_i s\alpha_i & -L_i s\beta_i c\alpha_i \\ c\beta_i s\alpha_i & L_i c\beta_i c\alpha_i & -L_i s\beta_i s\alpha_i \\ s\beta_i & 0 & L_i c\beta_i \end{bmatrix} \eta_i. \tag{3}$$

Specifically, by inversion (3), one have $\eta_i = J(\alpha_i, \beta_i)\dot{p}_i$, in which

$$J(\alpha_i, \beta_i) = \begin{bmatrix} c\beta_i c\alpha_i & c\beta_i s\alpha_i & s\beta_i \\ -s\alpha_i/L_i c\beta_i & c\alpha_i/L_i c\beta_i & 0 \\ -s\beta_i c\alpha_i/L_i & -s\beta_i s\alpha_i/L_i & c\beta_i/L_i \end{bmatrix}.$$

After taking the time derivative of η_i, we will have $\dot{\eta}_i = \dot{J}(\alpha_i, \beta_i)\dot{p}_i + J(\alpha_i, \beta_i)\ddot{p}_i$, then by multiply both sides of the obtained equation with $\bar{M}_i$ and substitute (2) to obtain

$$\bar{M}_i\dot{J}(\alpha_i, \beta_i)\dot{p}_i + \bar{M}_i J(\alpha_i, \beta_i)\ddot{p}_i + \bar{D}_i J(\alpha_i, \beta_i)\dot{p}_i = \tau_i. \tag{4}$$

Next, pre-multiply (4) by $J^T(\alpha_i, \beta_i)$, the following Euler-Lagrange-like dynamic model is derived for the target system

$$M_i(p_i)\ddot{p}_i + C_i(p_i, \dot{p}_i)\dot{p}_i + D_i(p_i)\dot{p}_i = \tau_i, \tag{5}$$

in which, $M_i(p_i) = J^T\bar{M}_i J$, $C_i(p_i, \dot{p}_i) = J^T\bar{M}_i\dot{J}$, $D_i(p_i) = J^T\bar{D}_i J$, $\tau_i = J^T\bar{\tau}_i$.

In model (5), $p_i \in \mathbb{R}^3$ indicates the generalized position of the ith UAV, $M_i \in \mathbb{R}^{3\times3}$ represents a positive-definite inertia matrix, $C_i \in \mathbb{R}^{3\times3}$ denotes the matrix of centripetal and Coriolis torque, $D_i \in \mathbb{R}^3$ incorporates the friction terms, and $\tau_i \in \mathbb{R}^3$ stands for a vector of the control inputs.

2) UAV Model: The kinematic model of the ith UGV is described as follows[37]

$$\begin{cases} \dot{x}_{ci} = v_i \cos\theta_i \\ \dot{y}_{ci} = v_i sin\theta_i \\ \dot{\theta}_i = w_{\theta i}, \end{cases} \tag{6}$$

where $p_{ci} = [x_{ci}, y_{ci}]^T \in \mathbb{R}^2$ denotes the position and θ_i denotes the orientation of the ith UGV, respectively, v_i and $w_{\theta i}$ represent the linear and angular velocity variables of the ith UGV, respectively. Similarly, the dynamic model of the UGV can also be formulated as in (2).

Since the nonholonomic constraint of the control system (6), to address this underactuated problem, similar to the "hand point" H_i used in UAV models, a "head point" H_i is introduced to transform the dynamic model of the ith UGV, and we then have its position coordinate $p_i = p_{ci} + L_i[cos\theta_i, sin\theta_i]^T$, and the corresponding time derivative

$$\dot{p}_i = \begin{bmatrix} \cos\theta_i & -L_i \sin\theta_i \\ \sin\theta_i & L_i \cos\theta_i \end{bmatrix} \eta_i. \tag{7}$$

Similarly, by inversion (7), one have $\eta_i = J(\theta_i)\dot{p}_i$, in which

$$J(\theta_i) = \begin{bmatrix} \cos\theta_i & \sin\theta_i \\ -\sin\theta_i/L_i & \cos\theta_i/L_i \end{bmatrix}.$$

After taking the time derivative of $\eta_i = J(\theta_i)\dot{p}_i$, then one have $\dot{\eta}_i = \dot{J}(\theta_i)\dot{p}_i + J(\theta_i)\ddot{p}_i$, premultiplying the resulting equation by $\bar{M}_i$, and substitute it the UGV dynamic model $\bar{M}_i\dot{\eta}_i + \bar{D}_i\eta_i = \bar{\tau}_i$, we obtain

$$\bar{M}_i J(\theta_i)\dot{p}_i + \bar{M}_i J(\theta_i)\ddot{p}_i + \bar{D}_i J(\theta_i)\dot{p}_i = \bar{\tau}_i. \tag{8}$$

Now, premultiplying (8) by $J^T(\theta_i)$, we can obtain the same Euler-Lagrange-like dynamic model of UAG as Eq. (5), that is $M_i(p_i)\ddot{p}_i + C_i(p_i, \dot{p}_i)\dot{p}_i + D_i(p_i)\dot{p}_i = \tau_i$.

It should be noted that system (5) possesses the following well-known properties[56,57].

Property 1: The mass matrix $M_i(p_i)$ is symmetric and uniformly positive definite.

Property 2: The mass matrix $M_i(p_i)$ and Coriolis/centripetal matrix $C_i(p_i, \dot{p}_i)$ satisfy skew-symmetric relationship $\mu^T(\dot{M}_i(p_i) - 2C_i(p_i, \dot{p}_i))\mu = 0$ for any arbitrary vector $\mu \in \mathbb{R}^3$.

Property 3: The dynamics exhibit linearity linear in the parameters

$$M_i(p_i)\dot{\mu} + C_i(p_i, \dot{p}_i)\mu + D_i(p_i)\dot{p}_i = Y_i(p_i, \dot{p}_i, \mu, \dot{\mu})\phi_i. \tag{9}$$

where $Y_i(p_i, \dot{p}_i, \mu, \dot{\mu}) \in \mathbb{R}^{3\times 12}$ is a known regression matrix and is only dependent on state variables, and $\phi_i \in \mathbb{R}^{12}$ is a constant vector of the model parameters in agent systems and given by

$$\phi_i = \begin{bmatrix} \begin{bmatrix} m_i & I_{1i}/L_i^2 & I_{1i}/L_i^2 & (\bar{D}_i)_{11} \end{bmatrix}^T \\ \begin{bmatrix} (\bar{D}_i)_{12}/L_i & (\bar{D}_i)_{13}/L_i & (\bar{D}_i)_{21}/L_i & (\bar{D}_i)_{22}/L_i^2 \end{bmatrix}^T \\ \begin{bmatrix} (\bar{D}_i)_{23}/L_i^2 & (\bar{D}_i)_{31}/L_i & (\bar{D}_i)_{32}/L_i^2 & (\bar{D}_i)_{33}/L_i^2 \end{bmatrix}^T \end{bmatrix}^T.$$

Graph theory

In this work, we consider a heterogeneous multi-UAV–UGV system with N agents, where N denotes an arbitrary positive integer, indicating that the number of agents contained in the system can be extended without an explicit upper bound. Let the undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$ denote the information communication topology of the multi-UAV–UGV system, which consists of M UAVs and $N - M$ UGVs. In graph $\mathcal{G}$, $\mathcal{V} = \{1, 2, \cdots, n\}$ and $\mathcal{E} = \{(i, j) \mid i, j \in \mathcal{V}, i \neq j\} \subseteq \mathcal{V} \times \mathcal{V}$ are the vertex set and edge set, respectively. The weighted adjacency matrix $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N\times N}$ with nonnegative elements denotes the graph adjacency structure[35,58]. An edge $(i, j) \in \mathcal{E}$ represents that vertex i is able to receive information from vertex j. For adjacency matrix $\mathcal{A}$, the weight $a_{ij} = 1$ if $(i, j) \in \mathcal{E}$, and $a_{ij} = 0$, otherwise. Meanwhile, we suppose $a_{ii} = 0$ for all $i \in \{1, 2, \cdots, N\}$, the set of neighbors of the ith vertex is defined by $\mathcal{N}_i(\mathcal{E}) = \{j \in \mathcal{V} \mid (i, j) \in \mathcal{E}\}$.

A framework is an implementation of a graph at specified points in $\mathbb{R}^d$, where $d \in \{2, 3\}$. If $p_i \in \mathbb{R}^d$ is the position of vertex i, then a d-dimensional formation framework $\mathcal{F}$ is represented as $(\mathcal{G}, p)$, where $p = [p_1, p_2, \cdots, p_N] \in \mathbb{R}^{dN}$. On the basis of an arbitrary ordering of graph $\mathcal{G}$, the edge function $f_{\mathcal{G}} : \mathbb{R}^{dN} \to \mathbb{R}^{|\mathcal{E}|}$ associated with $(\mathcal{G}, p)$ is defined as[56]

$$f_{\mathcal{G}}(p) = (\cdots, ||p_i - p_j||^2, \cdots), \quad (i, j) \in \mathcal{E}. \tag{10}$$

The rigidity matrix $R(p) : \mathbb{R}^{dN} \to \mathbb{R}^{|\mathcal{E}|\times dN}$ of a framework $(\mathcal{G}, p)$ is described as

$$R(p) = \frac{1}{2}\frac{\partial f_{\mathcal{G}}(p)}{\partial p}. \tag{11}$$

It is a well-known fact that if and only if $|\mathcal{E}| = 2|\mathcal{V}| - 3$ or $|\mathcal{E}| = 3|\mathcal{V}| - 6$, then a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$ is minimally rigid in two-dimensional (2-D) or 3-D respectively[59]. Thus, the rigidity matrix $R(p)$ has full row rank. Given that the matrix $R^T(p)$ has full column rank, as well as $\mathrm{rank}[R(p)] = \mathrm{rank}[R(p)R^T(p)]$, then $R(p)R^T(p) \in \mathbb{R}^{|\mathcal{E}|\times|\mathcal{E}|}$ is nonsingular (and hence invertible)[60].

Lemma 1 [52]: Let $\mu(t) \in \mathbb{R}^d$, and 1_N be the $N \times 1$ vector of ones, then $R(p)(1_N \otimes \mu) = 0$. We assume an undirected and connected communication graph during execution; the analysis can be extended to switching topologies that are jointly connected over finite time windows. Agents exchange neighboring agents' positions and velocities and the scalar v_d; localization is via onboard sensing (e.g., GNSS/INS and range sensors).

Artificial potential field

In this work, the APF algorithm is employed to plan the motion trajectory in the dense obstacle environments, aiming to achieve collision avoidance for the heterogeneous multi-UAV–UGV systems, the main reasons are summarized as follows: (1) It explicitly accounts for the overall characteristics of the formation. (2) It can handle trajectory planning problems in complex environments. (3) It has good real-time performance and flexibility. (4) It incorporates both local and global information. The fundamental idea of APF is that obstacles in the environment generate repulsive forces to push agents away, while the target generates an attractive force to guide the agents toward it. In the resultant field of repulsive and attractive forces, the agent moves in the direction corresponding to the minimum total potential energy.

The traditional attractive potential field and repulsive potential field, together with their corresponding potential forces, are given by the following functions[42].

$$\begin{cases} U_{att}(p_v) = \frac{1}{2}k_{att}d^2(p_v, p_{goal}) \\ F_{att}(p_v) = -\nabla U_{att}(p_v), \end{cases} \tag{12}$$

and

$$\begin{cases} U_{rep}(p_v) = \frac{1}{2}k_{rep}\left(\frac{1}{d(p_v, p_{obs})} - \frac{1}{\rho_0}\right)^2 \\ F_{rep}(p_v) = -\nabla U_{rep}(p_v), \end{cases} \tag{13}$$

where p_v represents the current waypoint planned by the leader UGV in the formation, which we call the virtual leader. $U_{att}(p_v)$ denotes the attractive potential field function formed by the goal point and $F_{att}(p_v)$ is the negative gradient of $U_{att}(p_v)$, i.e., the attractive force, whose direction points from the waypoint location toward the goal location. Similarly $U_{rep}(p_v)$ denotes the repulsive potential field function and $F_{rep}(p_v)$ represents the

negative gradient of $U_{rep}(p_v)$, i.e., the repulsive force, whose direction points from the obstacle location toward the agent. k_{att} and k_{rep} denote the positive attraction and repulsion coefficients, respectively. $d(p_v, p_{goal})$ indicates the relative distance from the waypoint p_v to the goal point p_{goal}, and $d(p_v, p_{obs})$ represents the relative distance from the waypoint p_v to the obstacle location p_{obs}. In (13), ρ_o is the influence radius of the obstacle repulsive force.

Problem formulation and control objective

Let the desired formation of heterogeneous multi-UAV–UGV systems be described by an infinitesimally and minimally rigid graph framework $\mathscr{F}^*(t) = (\mathscr{G}^*, p^*(t))$, where $\mathscr{G}^* = (\mathscr{V}^*, \mathscr{E}^*, \mathscr{A})$, $\dim(\mathscr{V}^*) = n$, $\dim(\mathscr{E}^*) = l$, and $p^* = [p_1^*, p_2^*, \cdots, p_N^*]$. Let the desired time-varying distance between the *i*th vehicle and the *j*th vehicle in the rigid framework be given by

$$d_{ij}(t) = ||p_i^*(t) - p_j^*(t)|| > d_{\text{safe}}, i, j \in \mathscr{V}^*, \tag{14}$$

where d_{safe} represents the minimum safe distance to avoid collisions between agents in the formation at any time, here, d_{safe} is chosen as twice the turning radius of the UAV.

In this article, we pay attention to the issue of formation maneuvering and motion planning. Given the real formation framework of UAVs and UGVs $\mathscr{F}(t) = (\mathscr{G}^*, p(t))$, where $p = [p_1, p_2, \cdots, p_N]$. It can be assumed that the relative location of vehicle pairs in $\mathscr{E}^*$ and the speed of each vehicle can be detected by onboard sensors. At the same time, we assume that at $t = 0$, $||p_i(0) - p_j(0)|| \neq d_{ij}(0), i, j \in \mathscr{V}^*$. The formation control objective is to design the control input $\tau_i = \tau_i(p_i - p_j, d_{ij}, d_{ij}, v_i - v_j, v_i)$ such that the multi-UGV–UAV system (5) satisfies the following conditions:

$$\|p_i(t) - p_j(t)\| - d_{ij}(t) \to 0, \text{ as } t \to \infty, \ i, j \in \mathscr{V}^*. \tag{15}$$

Remark: We adopt a distance-based specification. Under an infinitesimally rigid graph, enforcing desired distances yields the target shape up to a global translation and rotation; thus the formation pattern illustrated in the figures is consistent with distance-only control.

During the formation movement and obstacle avoidance process, the second control objective is stated as follows.

$$v_i(t) - v_d(t) \to 0, \text{as } t \to \infty, i = 1, 2, \cdots, N. \tag{16}$$

Assumptions: (i) $\mathscr{G}^*$ is connected; (ii) $\|p_i(0) - p_j(0)\| \neq d_{ij}(0)$ for all edges $(i, j) \in \mathscr{E}^*$; (iii) $d_{ij}(t) > d_{safe}$ and obstacle separation conditions hold; (iv) model parameters are constant but unknown; (v) external disturbances are bounded.

In (16), v_d represents the desired maneuver speed for each vehicle and is assumed to be a bounded and continuous function that is available to (or broadcast among) all vehicles. This implies that, once the desired heterogeneous formation structure is achieved, the multi-UAV–UGV system is capable of moving in space as a virtual rigid body, even when performing pure translation and/or rotation maneuvers.

Formation controller design

Consider the heterogeneous formation of multi-UAV–UGV systems governed by the Euler-Lagrange-like dynamic model (5), which can be rewritten as

$$\begin{cases} \dot{p}_i = v_i \\ M_i(p_i)\dot{v}_i = \tau_i - C_i(p_i, \dot{p}_i)v_i - D_i(p_i)v_i, \end{cases} \tag{17}$$

where v_i denotes the *i*th agent's H_i point velocity in the earth-fixed coordinate frame. We then define the relative position between any two agents in the multi-UGV–UAV formation system as

$$\tilde{p}_{ij} = p_i - p_j, (i, j) \in \mathscr{E}^*. \tag{18}$$

Then, for every edge in the graph $\mathscr{G}$, the corresponding distance error can be denoted as

$$e_{ij} = ||\tilde{p}_{ij}|| - d_{ij}, (i, j) \in \mathscr{E}^*. \tag{19}$$

By employing (17) and (19), the dynamics of distance error e_{ij} can be expressed as

$$\dot{e}_{ij} = \frac{d}{dt}\sqrt{\tilde{p}_{ij}^T\tilde{p}_{ij}} - \dot{d}_{ij} = (\tilde{p}_{ij}^T\tilde{p}_{ij})^{-\frac{1}{2}}\tilde{p}_{ij}^T(v_i - v_j) - \dot{d}_{ij} = \frac{\tilde{p}_{ij}^T(v_i - v_j)}{e_{ij} + d_{ij}} - \dot{d}_{ij}. \tag{20}$$

For the facilitation of subsequent introduction of the formation control method and stability analysis, we introduce the alternative error variable as follows[52].

$$\sigma_{ij} = ||\tilde{p}_{ij}||^2 - d_{ij}^2, (i, j) \in \mathscr{E}^*. \tag{21}$$

Using (19), (21) can be rewritten as

$$\sigma_{ij} = e_{ij}(||\tilde{p}_{ij}|| + d_{ij}) = e_{ij}(e_{ij} + 2d_{ij}). \tag{22}$$

Given that $||\tilde{p}_{ij}|| \neq d_{ij}$ because $||\tilde{p}_{ij}|| \geq 0$, it is straightforward to verify that $\sigma_{ij} = 0$ if and only if $e_{ij} = 0$. To facilitate the design of the formation control law and the subsequent stability analysis, we introduce the following potential function[46]

$$W_{ij}(e_{ij}) = \frac{1}{4}\sigma_{ij}^2 = \frac{1}{4}e_{ij}^2(e_{ij} + 2d_{ij})^2. \tag{23}$$

From the definition of W_{ij}, one know that it is positively definite and radially boundless in e_{ij} . Define

$$W(e) = \sum_{(i,j)\in\mathscr{E}^*} W_{ij}(e_{ij}), \tag{24}$$

where $e = [\cdots, e_{ij}, \cdots] \in \mathbb{R}^l$, and $W_{ij}(e_{ij})$ was given in (23). One can get the time derivative of (24) along (20)

$$\dot{W} = \sum_{(i,j)\in\mathscr{E}^*} e_{ij}(e_{ij} + 2d_{ij})[\tilde{p}_{ij}^T(v_i - v_j) - d_{ij}\dot{d}_{ij}]. \tag{25}$$

Using (11) and (22), Eq. (25) can also be denoted in the following form

$$\dot{W} = \sigma^T(R(p)v - d_v), \tag{26}$$

where $v = [v_1, v_2, \cdots, v_n] \in \mathbb{R}^{dN}$, $\sigma = [\cdots, \sigma_{ij}, \cdots] \in \mathbb{R}^l$, $d_v = [\cdots, d_{ij}\dot{d}_{ij}, \cdots] \in \mathbb{R}^l, (i,j) \in \mathscr{E}^*$, the elements of σ and d_v are arranged in the same order as those in (10).

Given that we are studying the Euler-Lagrange-like systems, based on the backstepping control technique, the Lyapunov function must be augmented accordingly. We therefore introduce the following variable.

$$s = v - v_f, \tag{27}$$

in which, v_f represents the fictitious velocity input at hand point H_i.

It should be noted that the controller design is premised on the assumption that the parameters ϕ_i in (9) are unknown. To address this problem, we define the parameter estimation error for the *i*th agent as

$$\tilde{\phi}_i = \hat{\phi}_i - \phi_i, \tag{28}$$

where $\hat{\phi}_i(t) \in \mathbb{R}^{12}$ is the dynamic estimation of ϕ_i.

We then update the Lyapunov function and obtain following function

$$W_d(e, s, \tilde{\phi}) = W(e) + \frac{1}{2}s^T M(p)s + \frac{1}{2}\tilde{\phi}^T\Gamma^{-1}\tilde{\phi}, \tag{29}$$

where W was defined in (24), $\Gamma \in \mathbb{R}^{12N\times 12N}$ represents a diagonal, positive-definite, and constant matrix, and it is assumed that $\dot{\phi} = 0$. Accordingly, the time derivative of the function (29) can be obtained

$$\begin{aligned}\dot{W}_d &= \dot{W}(e) + \frac{1}{2}(\dot{s}^T M(p)s + s^T\dot{M}(p)s + s^T M(p)\dot{s}) + \tilde{\phi}^T\Gamma^{-1}\dot{\tilde{\phi}} \\ &= \sigma^T(R(p)v - d_v) + \frac{1}{2}(\dot{s}^T M(p)s + s^T\dot{M}(p)s + s^T M(p)\dot{s}) + \tilde{\phi}^T\Gamma^{-1}\dot{\tilde{\phi}}.\end{aligned} \tag{30}$$

Since $\dot{s}^T M(p)s = s^T M(p)\dot{s}$ and based on Property 2, using (17), (27) and Property 3, one can rewrite (30) as

$$\begin{aligned}\dot{W}_d &= \sigma^T[R(p)(s + v_f) - d_v] + s^T(M(p)\dot{s} + C(p,\dot{p})s) + \tilde{\phi}^T\Gamma^{-1}\dot{\tilde{\phi}} \\ &= \sigma^T(R(p)v_f - d_v) + \sigma^T R(p)s + s^T[\tau - C(p,\dot{p})v + C(p,\dot{p})s - D(p)v - M(p)\dot{v}_f] + \tilde{\phi}^T\Gamma^{-1}\dot{\tilde{\phi}} \\ &= \sigma^T(R(p)v_f - d_v) + s^T[\tau - (M(p)\dot{v}_f + C(p,\dot{p})v_f + D(p)v) + R^T(p)\sigma] + \tilde{\phi}^T\Gamma^{-1}(\dot{\tilde{\phi}} - \dot{\phi}) \\ &= \sigma^T(R(p)v_f - d_v) + s^T[\tau - Y(p,\dot{p},v_f,\dot{v}_f)\phi + R^T(p)\sigma] + \tilde{\phi}^T\Gamma^{-1}\dot{\tilde{\phi}},\end{aligned} \tag{31}$$

where $\tau = [\tau_1, \tau_2, ..., \tau_N]$ denotes the constructed control input, which needs to adhere to the prescribed formation maneuvering conditions to render $\dot{W}_d$ negative semi-definite.

The following theorem establishes the formation control law for the heterogeneous multi-UAV–UGV formation maneuvering problem.

Theorem 1 Consider the heterogeneous multi-UAV–UGV formation framework $\mathscr{F}(t) = (\mathscr{G}^*, p(t))$ in $\mathbb{R}^3$. Then, the following proposed control input τ can ensure the asymptotic convergence of the formation error.

$$\tau = -k_a s + Y(p, \dot{p}, v_f, \dot{v}_f)\hat{\phi} - R^T(p)\sigma, \tag{32a}$$

$$v_f = R^+(p)(-k_v\sigma + d_v) + (1_N \otimes v_d), \tag{32b}$$

$$\dot{\hat{\phi}} = -\Gamma Y^T(p, \dot{p}, v_f, \dot{v}_f)s, \tag{32c}$$

in which, $v_d = (v_{d1}, v_{d2}, ..., v_{dN}) \in \mathbb{R}^{dN}$ (see (16)) is a bounded and continuous function defining the predetermined velocity for the formation agents, $R^+(p) = R^T(p)[R(p)R^T(p)]^{-1}$ denotes the Moore-Penrose pseudoinverse[61], and $k_a > 0$, $k_v > 0$, yields $(e, s) = 0$ proportionally stable and guarantees that (15) and (16) are met.

For each agent *i*, the distributed form is

$$\tau_i = -k_a s_i + Y_i(p_i, \dot{p}_i, v_{f,i}, \dot{v}_{f,i})\hat{\phi}_i - [R^T(p)\sigma]_i, \tag{33}$$

$$v_{f,i} = [R^+(p)(-k_v\sigma + d_v)]_i + v_d, \tag{34}$$

$$\dot{\hat{\phi}}_i = -\Gamma_i Y_i^T(p_i, \dot{p}_i, v_{f,i}, \dot{v}_{f,i})s_i. \tag{35}$$

Proof Without loss of generality, the stability of the designed controller can be verified via Lyapunov stability analysis. First of all, since the wanted formation framework $\mathscr{F}^*$ and the actual formation framework $\mathscr{F}(t)$ have inevitably the same edge numbers, for all $t \geq 0$, the minimal rigidity of $\mathscr{F}^*$ means that $\mathscr{F}(t)$ is also minimally rigid. Substituting (32a)–(32c) into (31), one can get □

$$\dot{W}_d = -k_v\sigma^T\sigma - k_a s^T s + R(p)(1_N \otimes v_d).$$

From Lemma 1, it follows that $R(p)(1_N \otimes v_d) = 0$.

$$\dot{W}_d = -k_v\sigma^T\sigma - k_a s^T s = -4k_v W - k_a\|s\|^2. \tag{36}$$

where (23) and (24) were used. From (36), it can be inferred that $\dot{W}_d \leq 0$ holds for all $t \geq 0$. Consequently, the function $W_d(t)$ is non-increasing (either decreasing or remaining constant) over the interval $t \geq 0$. Since (34) is negative definite, the level sets of W_d are invariant. Additionally, if $\tilde{p}(0) \in \mathscr{S}$, then $\tilde{p}(t)$ remains within $\mathscr{S}$ for all $t \geq 0$. Therefore, under the condition of (36), the equilibrium $e = 0$ is exponentially stable when $\tilde{p}(0) \in \mathscr{S}$. This exponential stability result implies that $\|p_i(t) - p_j(t)\| \to d_{ij}(t)$ as $t \to \infty$ for all $(i, j) \in \mathscr{E}_g^*$. Finally, from Eq. (36), it can be readily observed that $\sigma(e) \to 0$ as $e \to 0$. Given the above stability result and the fact that *R*(*p*) is uniformly bounded, it follows that $v_f \to (1_N \otimes v_d)$ as $e \to 0$. In the presence of bounded w_i, standard robust adaptive control arguments yield global uniform ultimate boundedness (GUUB) of (*e*, *s*) with an ultimate radius proportional to $\max_i \bar{w}_i$. Since we have proved that $e(t) \to 0$ as $t \to \infty$ in the disturbance-free case, it follows that $\dot{p}_i(t) - v_d(t)$ converges to a neighborhood of the origin whose size scales with $\max_i \bar{w}_i$; when $w_i \equiv 0$, this neighborhood shrinks to zero.

Formation motion planning

Adaptive safe APF algorithm based motion planning

We now turn our attention to the issue of motion planning for multi-UAV–UGV formations in dense obstacle environments. We assume that the first UGV can comprehensively process the environmental information collected by other agents and plan a reasonable formation motion path according to the obstacles. For this purpose, we take the first UGV as the leader UGV and the remaining UGVs and UAVs are follower vehicles, and we define the relative position between the planned path point (also known as the virtual leader UGV) p_v and the leader UGV p_N (i.e., the trajectory tracking error) as *Coupled stability (planner + controller):* Let the planner be a Lipschitz map $v_d = \Pi(p_v)$ with Lipschitz constant L_Π and a first-order filter $\tau_p \dot{v}_d = -v_d + \Pi(p_v)$ (or a rate limiter). If the backstepping inner-loop has input-to-state stability (ISS) gain γ from v_d to (*e*, *s*), then for τ_p sufficiently large and L_Π bounded, the cascade is ISS and admits GUUB of (*e*, *s*); in the limit $\tau_p \to \infty$ the separated design recovers Theorem 1. In practice, we implement a rate limiter $\|\dot{v}_d\| \leq \dot{v}_{\max}$ to avoid high-frequency excitation from ASAPF updates. Relation to prior work: Our adaptive repulsive gain follows the spirit of[62,63] but differs by (i) explicit formation-footprint awareness during scaling, (ii) a bounded Widrow–Hoff update with rate limiting consistent with the inner-loop ISS property, and (iii) a safety-certified pause-and-reroute fallback mechanism.

Rigidity and pseudo-inverse regularization: To avoid singularities of *R*(*p*), we use a damped least-squares pseudo-inverse $R^+(p) \leftarrow R^T(p)[R(p)R^T(p) + \lambda I]^{-1}$ with small $\lambda > 0$ and enforce a rigidity margin by rejecting waypoints that would violate minimal rigidity counts in 3D. During aggressive scaling, we monitor $\sigma(e)$ and reduce the step size when $\det(RR^T)$ drops below a predefined threshold.

Safety fallback (pause and reroute): When the estimated minimum passage width w_{gap} along the planned direction is below the minimum scaled footprint w_{min}, we pause the formation (hold $v_d = 0$), expand a corridor via inflated obstacles, and re-run ASAPF on the corridor graph (e.g., Dijkstra over sampled waypoints). If no feasible segment exists, we increase the formation scale back and try an alternative homotopy direction.

$$e_m = p_v - p_n, \tag{37}$$

and let $v_v := \dot{p}_v$. Next, we plan the motion trajectory of the heterogeneous multi-UAV–UGV formation in dense obstacle environments based on the position p_v where the formation is located.

Although traditional APF sometimes achieves satisfactory results in motion planning, its wide application is limited due to some inherent limitations, including the local minimum problem and the issue of unreachable targets. Specifically, when the attractive and repulsive forces balance at certain positions, the robot may become trapped in a local minimum and fail to reach the desired goal. Moreover, if obstacles are located near the target point, the repulsive force may dominate the attractive force, thereby preventing the robot from approaching the target. In complex environments, the design of suitable potential field functions becomes increasingly challenging, making these issues more likely to occur. The essential reason is that the goal point does not constitute the global minimum of the overall potential field, therefore, it is necessary to improve the traditional APF. An improved APF (IAPF) obstacle avoidance technique was proposed in[62] to obtain a better APF approach. In the proposed method, by bringing in the relative distance from the waypoint to the goal point, the traditional repulsive force potential field function is scaled by this distance factor, so that the resultant force acting on the agent vanishes at the goal point. As a result, the goal point remains the global minimum of the potential field, and the IAPF can also significantly improve the motion planning result. In the improved APF, the size of the attractive force potential field function from the goal and the corresponding attractive force function are redefined as

$$U_{att}(p_v) = \begin{cases} \frac{1}{2}k_{att}d^2(p_v, p_{goal}), & d(p_v, p_{goal}) \le \rho_g \\ k_{att}\rho_g d(p_v, p_{goal}) - \frac{1}{2}k_{att}\rho_g^2, & d(p_v, p_{goal}) > \rho_g, \end{cases} \tag{38}$$

$$F_{att}(p_v) = -\nabla U_{att}(p_v) = \begin{cases} -k_{att}d(p_v, p_{goal}), & d(p_v, p_{goal}) \le \rho_g \\ -k_{att}\rho_g, & d(p_v, p_{goal}) > \rho_g, \end{cases} \tag{39}$$

where ρ_g denotes the influence radius of the improved attractive potential field. When the distance from the waypoint p_v to the goal exceeds ρ_g, Eqs. (38) and (39) indicate that the growth rate of the attractive potential field slows down, while the attractive force remains unchanged. This modification addresses the issue of excessively strong attractive potentials when the UAV is far from the target. In scenarios where obstacles exist near the initial position, the attractive potential field is attenuated compared to the original model, whereas the attractive force remains unchanged. This adjustment enables smoother and more stable obstacle avoidance.

The improved repulsive potential field function from the obstacle is redefined as

$$U_{rep}(p_v) = \begin{cases} \frac{1}{2}k_{rep}\left(\frac{1}{d(p_v,p_{obs})} - \frac{1}{\rho_0}\right)^2 d^h(p_v, p_{goal}), & d(p_v, p_{obs}) \le \rho_0 \\ 0, & d(p_v, p_{obs}) > \rho_0, \end{cases} \tag{40}$$

where $d(p_v, p_{goal})$ denotes the distance from the waypoint p_v to the goal point p_{goal}, which is added to the repulsion field function as a distance dependent correction factor, and h is a positive constant.

When the formation of the multi-UAV–UGV system has not arrived at the goal point, i.e., $p_v \neq p_{goal}$, the repulsive force is given by

$$F_{rep}(p_v) = -\nabla U_{rep}(p_v) = \begin{cases} F_{rep1}(p_v)\vec{\mathrm{i}} + F_{rep2}(p_v)\vec{\mathrm{j}}, & d(p_v, p_{obs}) \le \rho_0 \\ 0, & d(p_v, p_{obs}) > \rho_0, \end{cases} \tag{41}$$

where $\vec{\mathrm{i}} = \nabla d(p_v, p_{obs})$ and $\vec{\mathrm{j}} = \nabla d(p_v, p_{goal})$ are two unit direction vectors, the former vector points from the obstacle toward the virtual leader UGV, while the latter vector points from the waypoint p_v toward the goal point, as illustrated in Fig. 3. It is clear that force $F_{rep1}(p_v)\vec{\mathrm{i}}$ repels the UGV away from obstacles, and force $F_{rep2}(p_v)\vec{\mathrm{j}}$ attracts the UGV toward the goal point.

For $0 < h < 1$, one can notice that the repulsive force potential field function (40) is not differentiable at the goal point p_{goal} when $d(p_v, p_{obs}) \le \rho_0$ and $d(p_v, p_{obs}) \neq 0$. When the formation of multi-UAV–UGV system lies within the sphere of influence of the obstacle potential field and has not arrived at the goal point, one can obtain

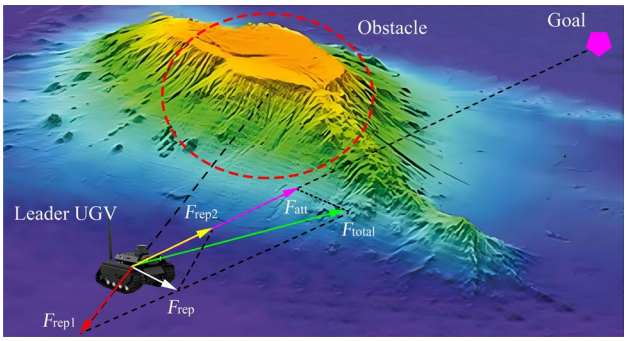


Fig. 3. The modified repulsive force model by the improved potential function.

$$F_{rep1}(p_v) = k_{rep}\left(\frac{1}{d(p_v,p_{obs})} - \frac{1}{\rho_0}\right)\frac{d^h(p_v,p_{goal})}{d^2(p_v,p_{obs})}\frac{\partial d(p_v,p_{obs})}{\partial(p_v)}, \tag{42}$$

$$F_{rep2}(p_v) = -\frac{h}{2}k_{rep}\left(\frac{1}{d(p_v,p_{obs})} - \frac{1}{\rho_0}\right)^2\frac{1}{d^{1-h}(p_v,p_{goal})}\frac{\partial d(p_v,p_{goal})}{\partial(p_v)}. \tag{43}$$

For $h = 1$, $d(p_v, p_{obs}) \le \rho_0$, and $d(p_v, p_{goal}) \neq 0$, (42) and (43) become

$$F_{rep1}(p_v) = k_{rep}\left(\frac{1}{d(p_v,p_{obs})} - \frac{1}{\rho_0}\right)\frac{d(p_v,p_{goal})}{d^2(p_v,p_{obs})}, \tag{44}$$

$$F_{rep2}(p_v) = -\frac{1}{2}k_{rep}\left(\frac{1}{d(p_v,p_{obs})} - \frac{1}{\rho_0}\right)^2. \tag{45}$$

From (44) and (45), we know that when the formation is close to the goal point, the first constituent part of the repulsive force $F_{rep1}(p_v)\bar{\mathrm{i}}$ approaches zero, while the second constituent part $F_{rep2}(p_v)\bar{\mathrm{j}}$ approaches a constant, thereby driving the formation toward the goal point.

For $h > 1$, one can see that (42) is differentiable at the goal point p_{goal}, and as the formation approaches the goal point, $d^h(p_v, p_{goal})$ and $d^{h-1}(p_v, p_{goal})$ approach zero. $F_{rep1}(p_v)\bar{\mathrm{i}}$, $F_{rep2}(p_v)\bar{\mathrm{j}}$ and the total repulsive force $F_{rep}(p_v)$ continuously approach zero.

Parameters tuning

Based on the IAPF method, although it can ameliorate the issues of local minima and unreachable targets to a certain extent, the problem of oscillations near dynamic obstacles still exists. In light of this, inspired by[63], we introduce the concept of adaptive safety on the basis of IAPF and design an enhanced adaptive safe artificial potential field (ASAPF) method to study the motion planning problem of heterogeneous multi-UAV–UGV systems in complex environments.

The selection of attraction and repulsion coefficients is critical for the normal operation of the formation, the improper parameter choices may prevent the formation from reaching the target point or even lead to collisions. To address this, an adaptive scaling coefficient strategy is proposed to ensure that the algorithm can plan feasible trajectories within a range suitable for the formation's footprint size. The tuning of the attractive and repulsive coefficients in the ASAPF framework is guided by a set of critical safety-oriented operating conditions that ensure reliable and collision-free formation motion. Specifically, two representative scenarios are considered. The first scenario focuses on the terminal phase of goal convergence, where smooth deceleration is required to determine the adaptive scaling factor of the attractive potential. The second scenario concerns obstacle avoidance and is characterized by the situation in which the distance between the planned waypoint and an obstacle equals the predefined safety threshold d_{safe}. At this critical distance, safety constraints dictate that, under the maximum allowable attractive force, the generated waypoint must induce a turning maneuver executed at the maximum admissible linear velocity.

To derive the attractive potential scaling under goal-reaching conditions, the following assumptions are introduced: (a) the planned formation maneuver waypoint and the target are aligned along the longitudinal direction, with their lateral (y-axis) coordinates being identical; (b) the Euclidean distance $d(p_v, p_{goal})$ between the planned waypoint p_v and the goal p_{goal} equals the prescribed goal margin distance; and (c) the environment is free of obstacles. Under these assumptions, the total artificial potential reduces to the attractive component alone, which enables a direct analysis of the attractive force behavior under this specific operating condition.

$$U^x_{total}(p_v) = U^x_{att}(p_v) = \frac{1}{2}k_{att}d^2(p_v,p_{goal}), \tag{46}$$

$$F^x_{total}(p_v) = F^x_{att}(p_v) = -k_{att}d(x_{goal},x_v), \tag{47}$$

where $(\cdot)^x$ denotes the x-component of the position vector, and x_v and x_{goal} are the x-coordinates of the planned waypoint p_v and the goal p_{goal}, respectively. Note that $d(x_v, x_{goal})$ represents the longitudinal distance between the planned waypoint and the goal. The key question is: what value of the force $F^x_{total}(p_v)$ at the critical distance ρ_g is needed to enforce a deceleration equal to a_v ? By invoking the braking-distance relation for uniformly decelerated motion, we obtain the following expression:

$$v_v = \sqrt{2a_v\rho_g}, \tag{48}$$

where v_v and a_v represent the velocity and linear deceleration of the planned waypoint. By substituting the distance expression ρ_g from (47) into (48), the scaling coefficient of attraction (k_{att}) can be explicitly derived as follows:

$$k_{att} = \frac{\sqrt{2a_v\rho_g}}{\rho_g}. \tag{49}$$

Next, the second critical operating condition is examined, corresponding to a fallback maneuver executed at the allowable formation motion velocity when the distance to an obstacle reaches the safety threshold d_{safe}. To

facilitate the analysis, the following assumptions are made: (a) the distance to the goal exceeds the predefined margin ρ_g; (b) a single obstacle is located directly ahead of the waypoint at a distance equal to d_{safe}; and (c) the planned waypoint, the goal, and the obstacle are aligned along the longitudinal direction, i.e., their lateral (y-axis) coordinates are all set to zero. Under these conditions, the x-component of the gradient of the total artificial potential can be expressed as:

$$F^x_{total}(p_v) = F^x_{att}(p_v) + F^x_{rep}(p_v). \tag{50}$$

Asuming that $d(p_v, p_{goal}) > \rho_g$, then according to (39), the attractive force is as follows

$$F^x_{att}(p_v) = -k_{att}\rho_g. \tag{51}$$

According to (41), the correspondingly repulsive force is as follows

$$F^x_{rep}(p_v) = k_{rep}\left(\frac{1}{d_{safe}} - \frac{1}{\rho_0}\right)\frac{d^h(x_v, x_{goal})}{d^2_{safe}}(x_v - x_{obs}) - \frac{h}{2}k_{rep}\left(\frac{1}{d_{safe}} - \frac{1}{\rho_0}\right)^2 d^{h-1}\frac{\partial d(x_v, x_{goal})}{\partial(p_v)}. \tag{52}$$

It is worth noting that the relative position satisfies $x_v - x_{obs} = -d_{safe}$. Furthermore, by enforcing the condition that the negative gradient of the total artificial potential equals the maximum admissible velocity v_v, the following expression is derived:

$$v_v = k_{rep}\frac{d_{safe} - \rho_0}{2d^2_{safe} \cdot {\rho_0}^2}\left(2\rho_0 d^h(x_v, x_{goal}) - h(d_{safe} - \rho_0)d^{h-1}\frac{\partial d(x_v, x_{goal})}{\partial(p_v)}\right) + \rho_g \cdot k_{att}. \tag{53}$$

Assuming that $x_{goal} > x_v$, (53) can be simplified to the following form:

$$v_v = k_{rep}\frac{d_{safe} - \rho_0}{2d^2_{safe} \cdot {\rho_0}^2}\left(2\rho_0 {\rho_g}^h + h{\rho_g}^{h-1}(d_{safe} - \rho_0)\right) + \rho_g \cdot k_{att}. \tag{54}$$

Adaptive scaling of potential field coefficients based on Widrow-hoff

The selection of the repulsive potential scaling coefficient (k_{rep}) is more complex. In this paper, the Widrow-Hoff rule is used to derive an adaptive equation for the repulsive potential scaling coefficient (k_{rep}), which was originally used to train neural networks with linear activation functions. Specifically, the Widrow-Hoff rule minimizes the mean squared error (E) between the desired value (y_d) and the output value (y) obtained from the current input ($\chi = (x_1, x_2, ..., x_m)$) and the corresponding input weights (w)

$$E(\mathrm{w}) = \frac{1}{2}e^2 = \frac{1}{2}(y_d - y)^2. \tag{55}$$

The weights get updated through the subsequent formula

$$w_j(k+1) = w_j(k) - \delta e(k)x_j(k). \tag{56}$$

where $w_j(k+1)$ and $w_j(k)$ represent the weights of the jth input at the $(k+1)$ and k iterations, respectively, and δ denotes the adaptation gain. The application of the Widrow-Hoff rule to the online adaptation problem of the repulsive potential scaling coefficient is given by the following update law.

$$k^{k+1}_{rep} = k^k_{rep} + \delta \cdot e_d \cdot d_{\min} \cdot \Delta t, \tag{57}$$

with

$$e_d = d_{safe} - d_{\min},$$

$$d_{\min} = \min\{d(p_v, p^i_{obs})\},$$

where k is the algorithm iteration, d_{min} is the minimum distance to obstacles, Δt indicates the time step, and δ is the adaptation gain. It is important to note that (47) must be bounded between prescribed lower and upper limits due to continuous-time integration. When the scaling factor is insufficient, collision-free operation cannot be guaranteed. Conversely, excessively large values may cause issues with goal-reaching performance. Therefore, based on the analytical derivation of the repulsive potential scaling factor proposed in[63], the boundary values are determined using the following criteria.

1) If the distance to the obstacle is the same as the safety margin (d_{safe}), the algorithm guarantees the movement velocity (v_v) in the opposite direction of the obstacle gradient.

$$k^{\min}_{rep} = f_{k_{rep}}(d_{safe}, -v_v). \tag{58}$$

2) If the distance to the obstacle is equal to the reaction-to-obstacle margin (ρ_0) subtract the safety margin d_{safe}, the algorithm ensures a movement velocity directed toward the goal position.

$$k_{rep}^{\max} = f_{k_{rep}}(\rho_0 - d_{safe}, v_v), \tag{59}$$

with

$$f_{k_{rep}}(d_{safe}, v_v) = \frac{2d_{safe}^2 \cdot {\rho_0}^2 \cdot (v_v - \rho_g \cdot k_{att})}{(d_{safe} - \rho_0)\left(2\rho_0 {\rho_g}^h + h{\rho_g}^{h-1}(d_{safe} - \rho_0)\right)}. \tag{60}$$

which comes from (54).

These defined limits for the repulsive potential scaling factor enable the suspension of adaptation when there are no obstacles in the robot's surroundings or when navigating narrow corridors where obstacles are close to the formation.

Calculation of the next waypoint

In the ASAPF, the total potential field force of the waypoint p_v in the formation is the sum of the attractive force from the goal point and the total repulsive force from obstacles and the goal point, that is

$$F_{total}(p_v) = F_{att}(p_v) + \sum_{i=1}^{K} F_{rep}(p_v), \tag{61}$$

in which, K represents the obstacle number in the environment.

In most cases, the resulting gradient of the potential function in the APF algorithm is considered as a velocity command. In such a case, the movement direction (θ) of the planned waypoint can be calculated using the following formula

$$\theta = \arctan(F_{total}^{y}(p_v)/F_{total}^{x}(p_v)), \tag{62}$$

where F_{tatal}^{x} and F_{tatal}^{y} denote the components of the resultant force F_{total} in the X-axis direction and Y-axis direction, respectively.

$$\begin{cases} x_v^{next} = x_v + v_v \cdot \Delta t \cdot \cos(\theta) \\ y_v^{next} = y_v + v_v \cdot \Delta t \cdot \sin(\theta), \end{cases} \tag{63}$$

where $p_v = (x_v, y_v)$ is the current waypoint position. After obtaining the current path point, it is necessary to predict whether the current path point needs to fallback based on the distance to the obstacle. if satisfied

$$\|p_v - p_{obs}\| \leq d_{safe}, \tag{64}$$

then it is determined to be unsafe and changed to a half-step advancement

$$\begin{cases} x_v^{next} = x_v + \frac{1}{2} v_v \cdot \Delta t \cdot \cos(\theta) \\ y_v^{next} = y_v + \frac{1}{2} v_v \cdot \Delta t \cdot \sin(\theta). \end{cases} \tag{65}$$

By iterating the above steps, all planned path points in the potential field can be obtained in turn; the smaller the calculation step size, the smoother the planned formation motion trajectory will be. Finally, all planned waypoints are connected to form a continuous path, that is, the planned obstacle avoidance path in the obstacle environment.

The below laws give the main result of this section.

$$\tau = -k_a s + Y(p, \dot{p}, v_f, \dot{v}_f)\hat{\phi} - R^T(p)\sigma, \tag{66a}$$

$$v_f = u_a + [1_N \otimes (v_v + k_m e_m)], \tag{66b}$$

$$u_a = R^{+}(p)(-k_v \sigma + d_v), \tag{66c}$$

$$\dot{\hat{\phi}} = -\Gamma Y^T(p, \dot{p}, v_f, \dot{v}_f)s, \tag{66d}$$

where $k_m > 0$. Additionally, regarding parameter sensitivity, k_a predominantly influences the convergence time of the formation. A larger k_a accelerates formation convergence but simultaneously increases the control input τ, thereby escalating energy consumption. A reasonable range for k_a is $2 < k_a < 8$. For k_v, it mainly affects the control input τ, a higher k_v value amplifies τ, potentially causing overshoot and affecting formation convergence time. An optimal interval for k_v is $0.5 < k_a < 1.5$. As for k_m, it primarily impacts the leader UGV's tracking of the planned trajectory. A larger k_m extends the formation's convergence time, with a recommended range of $0 < k_m < 1$. Then, (66) renders $e \to 0$ and $s \to 0$ asymptotically stable, and ensures that (15) and (16) are

met. At the same time, the heterogeneous multi-UGV–UAV systems realize formation maneuvering and motion planning in dense obstacle environments.

Input: $p_{v,0}$, p_g, $\{p_{obs}\}_{i=1}^{K}$, k_{att}, $k_{rep,0}$, ρ_g, ρ_0, d_{safe}, v_v, h
Output: Path $\mathscr{P}$

$p_v \leftarrow p_{v,0}$, $k_{rep} \leftarrow k_{rep,0}$, $\mathscr{P} \leftarrow \{p_{v,0}\}$
while $\|p_v - p_{goal}\| > \varepsilon_g$ **do**
  Compute $d_g \leftarrow \|p_v - p_{goal}\|$, $\hat{e}_g \leftarrow (p_v - p_{goal})/\|p_v - p_{goal}\|$
  if $d_g \le \rho_g$ **then**
    $F_{att} \leftarrow -k_{att} d_g \hat{e}_g$
  else
    $F_{att} \leftarrow -k_{att} \rho_g \hat{e}_g$
  end if
  $F_{rep} \leftarrow \mathbf{2}$
  for $i = 1$ to K **do**
    if $\|p_v - p_{obs,i}\| \le \rho_0$ **then**
      $\phi \leftarrow \frac{1}{d_{obs}} - \frac{1}{\rho_0}$
      $F_{rep} \leftarrow F_{rep} + k_{rep}\phi \frac{d_g}{d_{obs}^2}\hat{i} - \frac{1}{2}k_{rep}\phi^2 \hat{j}$
    end if
  end for
  $F_{total} \leftarrow F_{att} + F_{rep}$
  $k_{rep} \leftarrow \mathrm{clip}(k_{rep} + h(d_{safe} - \rho_0)\rho_0)$
  $\theta \leftarrow \arctan(F_{total}^y / F_{total}^x)$
  $p_v \leftarrow p_v + v_v \Delta t \theta$
  $\mathscr{P} \leftarrow \mathscr{P} \cup \{p_v\}$
end while
return $\mathscr{P}$

Algorithm 1. Adaptive ASAPF-Based Formation Maneuver Path Planning

Simulation results and analysis

This section presents a sequence of simulation results to verify the validity and effectiveness of the designed formation control and motion planning algorithms, and the experimental environment is the MATLAB 2024a simulation platform. We consider a heterogeneous multi-UAV–UGV formation system including four follower UAVs (indices 1 to 4), four follower UGVs (indices 5 to 8), and one leader UGV (index 9). The leader UGV is located at the geometric center of the UGV formation, and the communication interactions among the UAVs and UGVs are modeled by a 3-D undirected graph, as shown in Fig. 4, where the UGVs run on the ground, and the UAVs fly in the sky. We assume that each agent is equipped with reliable sensors (such as LiDAR, ultrasonic sensors, cameras, and infrared sensors)[64] that can measure the distance between itself and its neighboring agents to achieve distance-based formation control. In the formation structure, the UGVs on the ground are arranged in a square, the UAVs in the sky are arranged in a square at a prescribed altitude from the ground. These two squares together form a three-dimensional polyhedral structure, enabling formation coordination tasks in multi-obstacle environments.

Simulation parameter setting

The simulation aims to demonstrate the performance of the proposed algorithm in achieving formation motion planning and obstacle avoidance. We simulated some realistic scenarios, where the heterogeneous multi-UAV–UGV formation system navigates in the area through the ASAPF planned waypoints to perform surveillance,

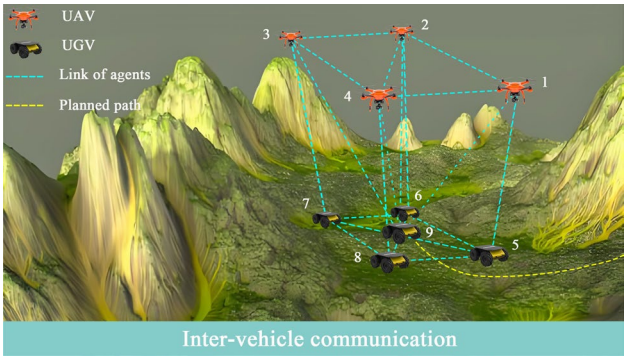


Fig. 4. Communication topology of the UAV–UGV systems.

Parameter	Description	Value [units]
m_i	Vehicle mass	3.5 [kg]
I_{1i}	Inertial moment 1	0.5000 [kg-m^2]
I_{2i}	Inertial moment 2	0.6000 [kg-m^2]
$\bar{D}_i$	Damping matrix	Diag{0.3, 0.004,0.5}
L_i	Length of vehicle hand (see Fig. 1)	0.2 [m]
h	Exponent in U_{rep}	1 [/] (chosen to balance repulsion and goal attraction; $h < 1$ weakens
		far-field repulsion, $h > 1$ strengthens)
δ	Adaptation gain	0.02 [/] (selected to satisfy $\|\dot{v}_d\| \leq \dot{v}_{\max}$ and monotone d_{min} in trials)
k_a	Control gain in (66a)	5 [/]
k_v	Control gain in (66c)	1 [/]
k_m	Control gain in (66b)	0.5 [/]
Γ_i	Definite matrix in (66d)	10I [/]
$\hat{\phi}_i(0)$	Initial parameter estimate	0 [/]

Table 4. Parameters for multi-UAV–UGV formation control and motion planning.

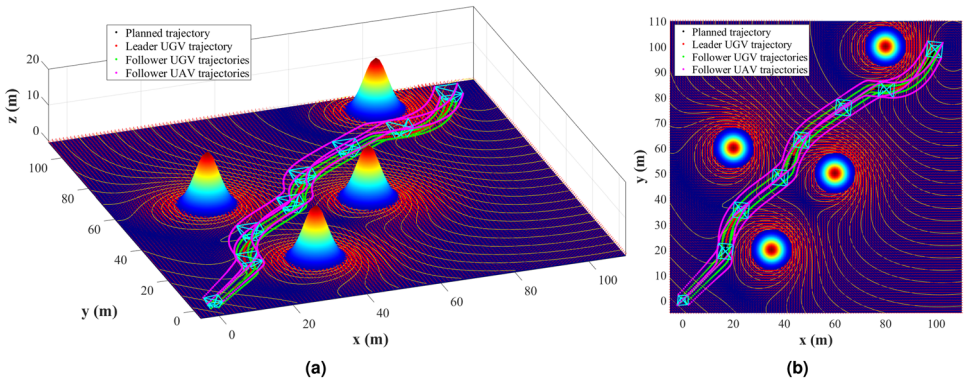


Fig. 5. Position trajectories of the heterogeneous multi-UAV–UGV formation system in an environment with sparse obstacles. **a** 3-D trajectories. **b** The 2-D trajectories ($X - Y$ plane view).

reconnaissance, or search tasks. To realize this, four scenarios were designed to exhibit the main challenges and variations that the multi-UAV–UGV formation systems may face. These scenarios are a sparse-obstacle scenario, a dense-obstacle scenario, a scenario with dynamic obstacles, and a scenario with more agents form a hexagonal prism. In the simulations, the leader UGV is set to plan the motion path in the obstacle environment, the follower UGVs track the leader UGV on the ground, and UAVs fly with the UGV at a fixed or time-varying altitude.

In all these scenarios, the ASAPF is established in the area of $65\ m \times 65\ m$, and the formation's initial position was set at (0, 0, 0), while the goal position was designated as (60, 60, 0), the starting position and end position of the leader UGV1 are set as (0, 0, 0) and (60, 60, 0), the starting position of the follower UGV5 is $(-1.0, -0.9)$, UGV6 is $(1.1, -1.2)$, UGV7 is (0.9, 0.8), UGV8 is $(-0.8, 0.8)$, and the UAVs are at a certain height from the UGV which are assumed to take off from their corresponding UGVs. The position value setup method is similar to that in[47], and Table 4 lists the parameters used in the simulations. Four different scenarios are shown in this section to compare the formation maintenance and obstacle avoidance resulting from the proposed control method in this paper. We define sparse versus dense obstacles by occupancy ratio and clearance: sparse if free-space occupancy $< 20\%$ and minimum gap $> 1.5\times$ formation diameter; dense if occupancy $\geq 40\%$ or minimum gap $\leq 1.2\times$ formation diameter. Scenario parameters follow Table 4.

Formation maneuvering in an environment with sparse obstacles

In the first scenario presented in this section, the heterogeneous multi-UAV–UGV formation system maneuvers in an outdoor environment with sparse obstacles, such as the UAV–UGV formation performing measurement work over a plain or carrying out patrol missions on the edge of the city. In such scenarios, the formation needs to maintain a specific structure and move along the planned trajectory to perform tasks, such as target search. Four static obstacles are considered, and their positions are $p_{obs1} = (8, 30, 0)$, $p_{obs2} = (16, 10, 0)$, $p_{obs3} = (32, 28, 0)$, and $p_{obs4} = (40, 50, 0)$, respectively. First, four UAVs fly to constant height ($h = 5$ m) and subsequently, four UAVs fly and keep a square formation along with the UGVs. Figure 5 indicates the simulation

results of the heterogeneous multi-UGV–UAV formation maneuvering strategy under the proposed rigid graph based formation control method in an environment with few obstacles, where Fig. 5a represents the 3-D position trajectories, while Fig. 5b denotes the 2-D ($X-Y$ plane) position trajectories. In Fig. 5, the yellow dotted line denotes the planned trajectory based on the ASAPF method, the red dotted line depicts the trajectory of the leader UGV, the green dotted lines depict the trajectories of the follower UGVs, and the pink dotted lines depict the trajectories of the follower UAVs. The sky blue polyhedral configurations represent the formation configuration every 20 seconds, showing that the UAV and UGV maintain the preset distance in both the horizontal and vertical directions, thereby verifying the validity of the distance consensus protocol under rigid graph theory. The thin solid yellow lines denote the equipotential lines of the APF, while the red dotted lines indicate the magnitude and distribution of the repulsive force from obstacles. In Fig. 5, we illustrate the formation acquisition and maneuver process of heterogeneous UAV–UGV formation with the trajectory of each agent. It can be clearly seen that under the designed control law (53), five UGVs and four UAVs converge asymptotically to the predefined formation configuration, the trajectory of each agent is smooth and closely follows the planned obstacle avoidance trajectory, the Formation maintains a stable 3-D structure during the movement process, and no collisions or trajectory divergence phenomena are observed.

Figure 6a gives the distance errors $e_{ij}(t), i, j \in \mathscr{V}^*$ among each agent in the formation, which rapidly converge to zero within 4 seconds, that the system possesses strong transient regulation capability. Moreover, a larger initial error leads to a longer adjustment period during the early stage of the response. Nevertheless, by integrating rigid graph theory with backstepping-based adaptive control, the system is still able to achieve rapid convergence, thereby demonstrating the robustness of the proposed control law with respect to large initial errors. Figure 6b illustrates the control inputs $\bar{\tau}_i$ for all formation agents along the three coordinate axes. These inputs fluctuate slightly in the initial stage and then gradually stabilize, which reflects the adaptability of the controller to model uncertainties. Figure 6c shows the parameter estimates $\hat{\phi}_1$ for follower UAV 1. The results clearly demonstrate that $\hat{\phi}_1$ gradually reaches constant values and stabilizes over time, thereby verifying the effectiveness of the adaptive mechanism. While Fig. 6d shows the adaptive variation curve of the repulsive force coefficient k_{rep}.

Formation scaling reconfiguration maneuvering in an environment with dense obstacles

This scenario simulates an outdoor environment with dense obstacles, such as the heterogeneous UAV–UGV formation performing target search and rescue tasks in the dense jungle of mountainous areas. In this example, compared to the previous simulations, we consider a formation scaling configuration scenario, where the original formation shapes need to be scaled to avoid large or dense obstacles in a more complex environment (The scaling

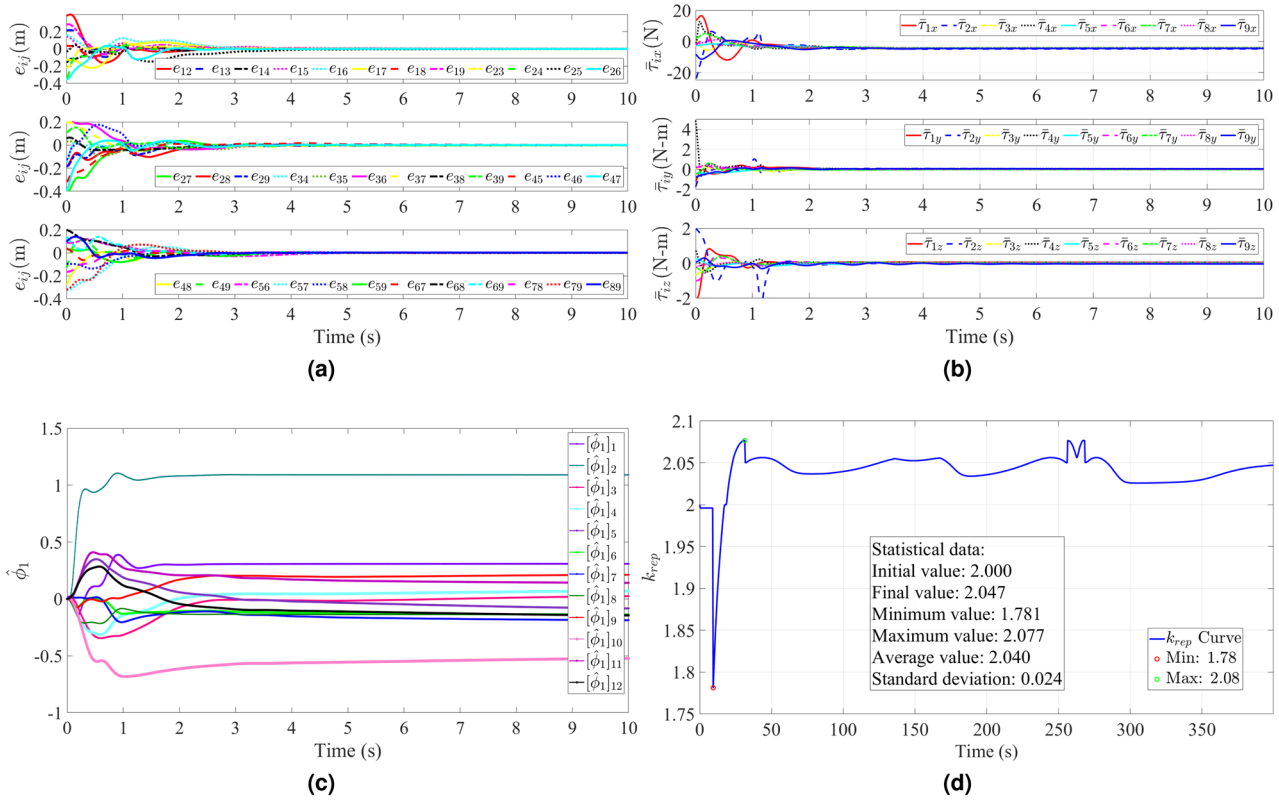


Fig. 6. **a** The distance errors $e_{ij}(t), i, j \in \mathscr{V}^*$ between any two agents. **b** The formation control input $\bar{\tau}_i$ for each agent in *x*, *y*, and *z* directions. **c** Sample of parameters estimates for follower UAV 1, $\hat{\phi}_1$. **d** Adaptive variation curve of the repulsive force coefficient k_{rep}.

of the formation here is achieved based on affine transformation. Intuitively, affine scaling preserves the relative geometric relationships among agents while uniformly adjusting the formation size, enabling coordinated expansion or contraction without altering the formation topology. The scaling factor is determined based on the size of the current formation configuration and the minimum distance to be passed through the channel. Please refer to our previous work, Reference[28]). In this scenario, we set 14 static obstacles, with their positions at p_{obs1} = (2,45,0), p_{obs2} = (7,30,0), p_{obs3} = (10,8,0), p_{obs4} = (15,55,0), p_{obs5} = (25,38,0), p_{obs6} = (28,5,0), p_{obs7} = (30,22,0), p_{obs8} = (30,50,0), p_{obs9} = (42,38,0), p_{obs10} = (42,56,0), p_{obs11} = (46,12,0), p_{obs12} = (50,26,0), p_{obs13} = (55,42,0), and p_{obs14} = (60,5,0). Similar to the previous simulation, in this scenario, the UGVs and UAVs also start from the starting position and form a heterogeneous formation of a specific size and shape after 5 s. However, as the formation moves, more and more obstacles are encountered, and the formation needs to adaptively change its size to pass through the dense obstacle environment. Figure 7a and b exhibit the obstacle avoidance results of a heterogeneous multi-UAV–UGV formation system using the flexible motion planning strategy (53) in an environment with dense obstacles, where Fig. 7a shows the 3-D trajectory and Fig. 7b shows the corresponding 2-D trajectory.

As can be seen from these two figures, when facing 14 dense obstacles, the heterogeneous multi-UAV–UGV formation crossed the narrow area by adaptively adjusting the formation size (i.e., scaling the formation configuration) and maintained a stable time-varying formation structure as expected, successfully avoiding the dense obstacles and reaching the destination, the 2-D trajectory shows that the inflection points of its path are smooth and free of redundant turns. The simulation results demonstrate that the designed motion planning strategy ensures the formation maintains an appropriate time-varying geometry and navigates an environment with dense obstacles without collision. Moreover, the dynamic adjustment of the ASAPF repulsive force factor enables the formation to maintain a safe distance in the obstacle cluster, and no stagnation caused by local minima is observed.

Figure 8a illustrates the inter-agent distance errors $e_{ij}(t)$, $i, j \in \mathcal{V}^*$ within the formation, while Fig. 8b depicts the control inputs $\bar{\tau}_i(t)$ for all formation agents along the three coordinate axes. It can be observed that the convergence time of the distance error is similar to that of the sparse scene (approximately 4 seconds), but the control input amplitude is slightly larger, reflecting the additional adjustment effort required under dense obstacles. Figure 8c presents the parameter estimates for follower UAV 2, clearly demonstrating that these estimates gradually converge to and stabilize at constant values over time. The parameter estimation stability is not affected by environmental complexity and remains within a bounded range. Meanwhile, Fig. 8d shows the adaptive variation curve of the repulsive force coefficient k_{rep}.

Formation maneuvering in an environment with dynamic obstacles

The simulation of formation maneuvers in an environment with dynamic obstacles, including vehicles and birds, can validate the effectiveness of the designed algorithm in addressing these challenges. In this simulation, a dynamic environment incorporating two static obstacles and two dynamic obstacles was constructed to validate the dynamic obstacle avoidance capability of the heterogeneous multi-UAV–UGV formation system, the static obstacles are positioned at coordinates $p_{obs1} = (16, 10, 0)$ and $p_{obs3} = (32, 28, 0)$, while the initial positions of the two spherical dynamic obstacles are set at $p_{obs3} = (0, 42, 0)$ and $p_{obs4} = (40, 50, 0)$, respectively. Specifically, dynamic obstacle 1 moves in a fixed direction at a velocity of $(0.25, -0.25, 0)$ m/s, while dynamic obstacle 2 executes circular motion with a velocity vector given by $(-3\sin(0.2t), 2\cos(0.2t), 0)$ m/s. The formation's initial positions, target positions, and other parameters remain identical to those specified in Part A of the simulation.

Figure 9 illustrates the maneuvering trajectories of the formation in a dynamic obstacle environment. Specifically, Fig. 9a displays the 3-D trajectory, while Fig. 9b shows the 2-D trajectory. These results demonstrate that, even in the presence of dynamic obstacles, the UAV–UGV formation maintains stable adherence to the desired formation. When encountering dynamic obstacles, the adjustment of the dynamic repulsive force factor, based on the ASAPF algorithm, enables the formation to plan a detour path in advance and quickly generate a safe and collision-free trajectory. The formation adjusts its heading and velocity in real time, successfully evading

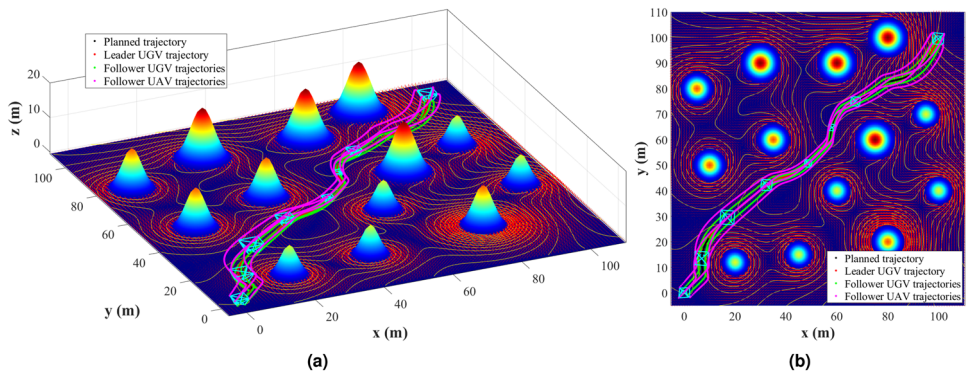


Fig. 7. Position trajectories of the heterogeneous multi-UAV–UGV formation system in an environment with dense obstacles. **a** 3-D trajectories. **b** The 2-D trajectories ($X - Y$ plane view).

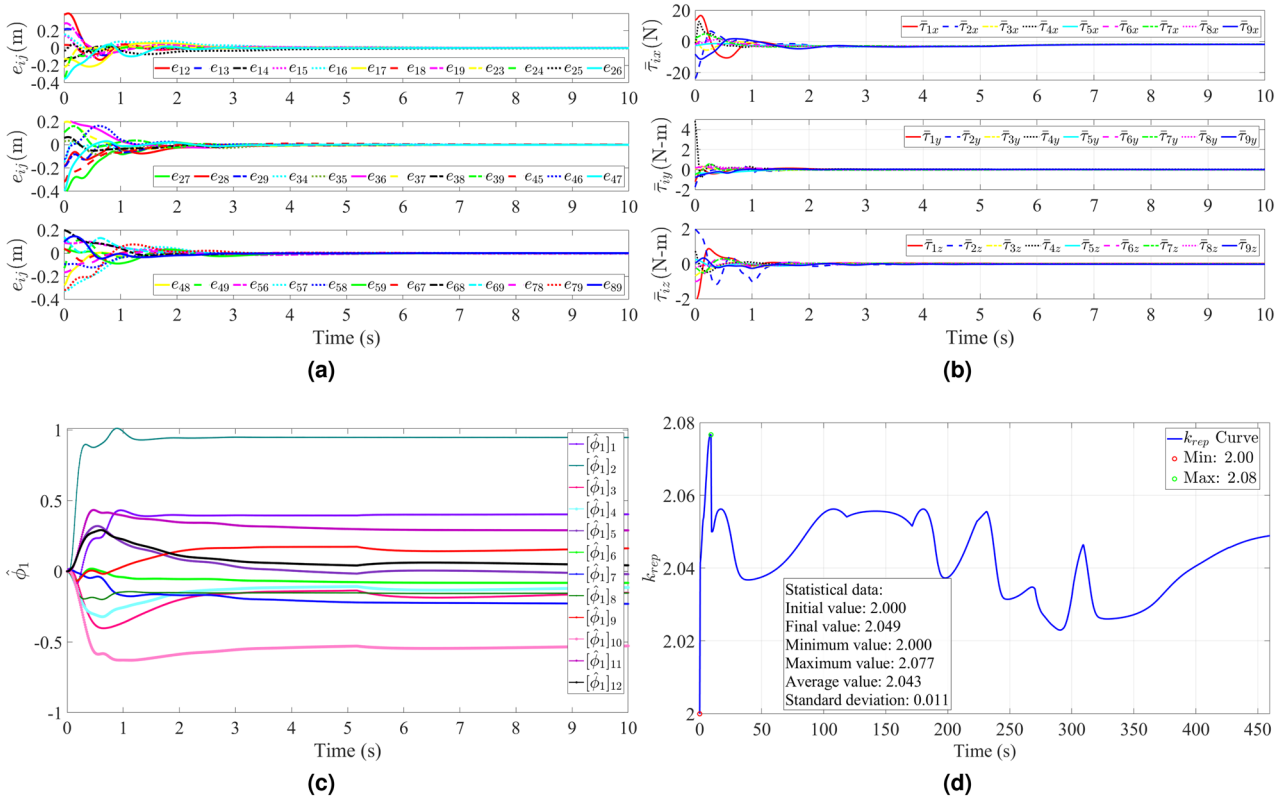


Fig. 8. **a** The distance errors $e_{ij}(t), i, j \in \mathcal{V}^*$ between any two agents. **b** The formation control input $\bar{\tau}_i$ for each agent in *x*, *y*, and *z* directions. **c** Sample of parameters estimates for follower UAV 2, $\hat{\phi}_2$. **d** Adaptive variation curve of the repulsive force coefficient k_{rep}.

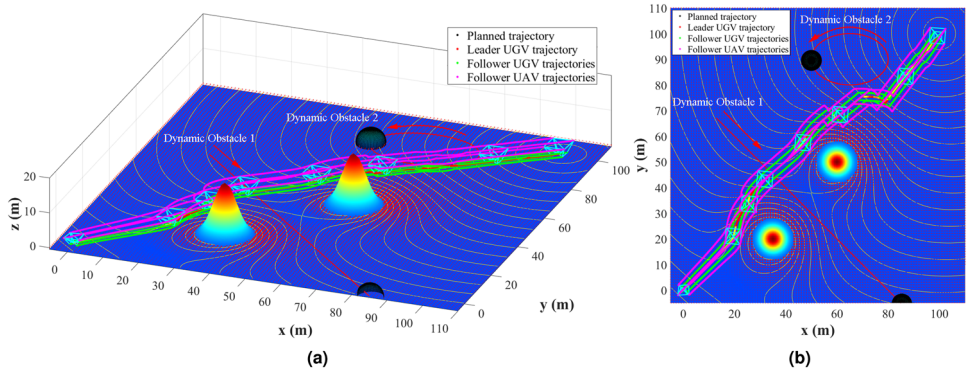


Fig. 9. Position trajectories of the heterogeneous multi-UAV–UGV formation system in an environment with dynamic obstacles. **a** 3-D trajectories. **b** 2-D trajectories ($X - Y$ plane view).

both types of obstacles. During the entire simulation process, the minimum distance between the formation and obstacles remains above the predefined safe threshold to ensure collision-free navigation. The 2D trajectory further shows that during the obstacle avoidance process, the formation did not interrupt the continuity of motion, and there is no severe oscillation at the path turning points, verifying the smoothness advantage of ASAPF and the effectiveness of the obstacle avoidance strategy in the face of dynamic obstacles.

Figure 10a gives the distance errors $e_{ij}(t), i, j \in \mathcal{V}^*$ between any two agents. Figure 10b illustrates the control inputs $\bar{\tau}_i(t)$ for agents in the direction of three coordinate axes, which exhibit an instantaneous peak when a dynamic obstacle is detected and then quickly returns to stability, reflecting the adaptive adjustment efficiency of the repulsive force factor under the Widrow-Hoff rule. Figure 10c presents the parameter estimation results for the leader UGV, indexed as 9, and Fig. 10d shows the adaptive variation curve of the repulsive force coefficient k_{rep}. During the process of obstacle avoidance in dynamic obstacles (t_1 = 35-60 s and t_2 = 155-190 s), the maximum tracking error reached 0.2 m, and it recovers to 0.1 m within 3 seconds after obstacle avoidance.

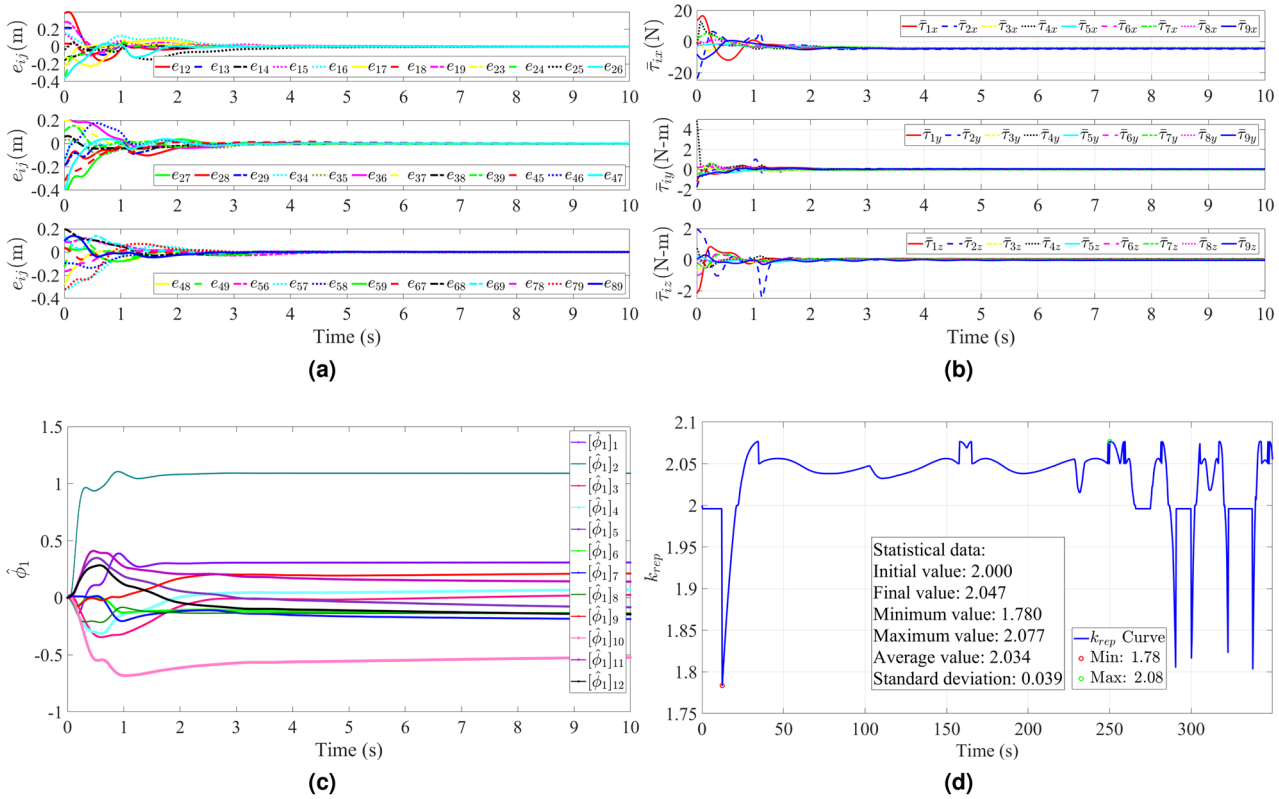


Fig. 10. **a** The distance errors $e_{ij}(t), i, j \in \mathcal{V}^*$ between any two agents. **b** The formation control input $\bar{\tau}_i$ for each agent in *x*, *y*, and *z* directions. **c** Sample of parameters estimates for leader UGV 9, $\hat{\phi}_9$. **d** Adaptive variation curve of repulsive force coefficient k_{rep}.

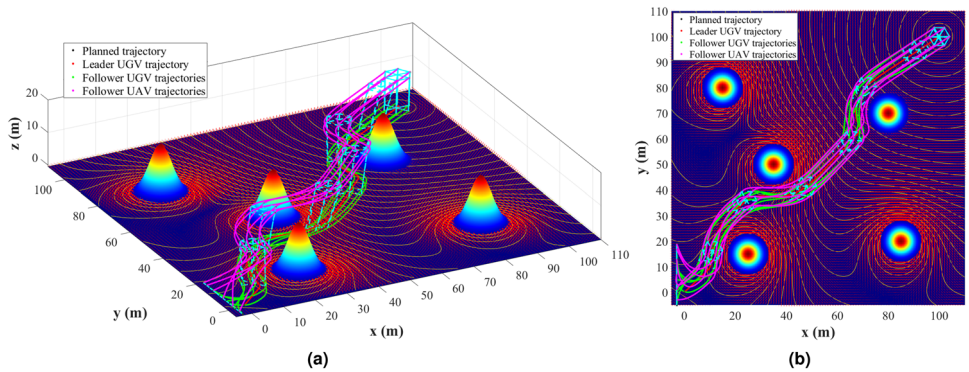


Fig. 11. Position trajectories of the heterogeneous multi-UAV–UGV formation system in hexagonal prism configuration. **a** 3-D trajectories. **b** 2-D trajectories ($X-Y$ plane view).

Cooperative maneuvering of heterogeneous multi-UAV–UGV formation in hexagonal prism configuration

In this simulation study, a total of 13 agents–comprising 7 UGVs and 6 UAVs are deployed, with the desired formation specified as a regular hexagonal prism. Figure 11 illustrates the cooperative maneuvering trajectories of the heterogeneous UAV–UGV formation in an environment containing five static obstacles, where Fig. 11a presents the 3-D view and Fig. 11b shows the corresponding 2-D projection.

As observed, all agents initially start from the same linear arrangement. During the maneuver, the UGVs autonomously organize into a regular hexagonal formation on the ground plane, while the UAVs ascend to form a second regular hexagon in the air. Together, these two planar formations constitute the prescribed hexagonal prism structure, which then proceeds smoothly toward the target location. Throughout the cooperative navigation process, the formation exhibits strong configuration preservation, effective obstacle avoidance, and

high path-tracking accuracy. These results demonstrate the suitability of the proposed approach for complex multi-robot cooperative missions, such as coordinated reconnaissance and area coverage tasks.

From the transient response perspective, the evolution of the state-related errors has a direct impact on the formation dynamics. As shown in Figs. 6a, 8a, and 10a, relatively large initial distance and velocity errors lead to noticeable transient fluctuations in the early stage, which are mainly caused by mismatches between the initial formation configuration and the desired rigid formation, as well as abrupt changes in the planned velocity induced by obstacle avoidance. These transient errors temporarily increase the corrective control effort, resulting in short-lived oscillations in both the inter-agent distance errors and control inputs. However, owing to the backstepping-based error dynamics and the damping effect introduced by the *s*-variable, all state errors are rapidly attenuated and converge to a small neighborhood of zero within a few seconds. This indicates that the proposed control framework effectively suppresses transient amplification caused by state deviations and ensures fast convergence without compromising formation stability.

Comparison with other algorithms

To further validate the superiority of the proposed RG-ASAPF algorithm in heterogeneous multi-UAV–UGV formation control and motion planning, a 3-D simulation environment with multiple obstacles was established. The performance of the designed RG-ASAPF method was compared with five representative algorithms under identical simulation conditions: the traditional APF-, an improved APF-, a PSO-, a MPC-, and a Nonlinear MPC (NMPC)-based methods. The simulation was conducted in a $110 \times 110 \times 20$ space containing nine obstacles, and their positions were respectively at $p_{obs1} = (22, 8, 0)$, $p_{obs2} = (30, 45, 0)$, $p_{obs3} = (5, 55, 0)$, $p_{obs4} = (70, 25, 0)$, $p_{obs5} = (15, 90, 0)$, $p_{obs6} = (80, 90, 0)$, $p_{obs7} = (90, 55, 0)$, $p_{obs8} = (50, 85, 0)$, and $p_{obs9} = (60, 65, 0)$. The formation was required to maintain structural stability while executing obstacle avoidance maneuvers.

The simulation results are presented in Fig. 12. Specifically, Fig. 12a illustrates the formation motion trajectories of multi-UAV–UGV system using the traditional APF based algorithm, which achieves global path optimization but produces relatively stiff trajectories at turns, and its prone to some local minima, leading to oscillations near obstacles. Figure 12b illustrates the formation motion trajectories using the improved APF based algorithm, which partially alleviates the oscillations through repulsive force adjustment. Figure 11c shows the formation motion trajectories of the PSO based algorithm. Although this method can identify a relatively optimal path, in complex obstacle environments, premature particle convergence of particles causes some agents trajectories collide with obstacles, preventing them from reaching the target position smoothly. Figure 12d shows the outcomes of the MPC based algorithm. Due to the need for continuous prediction and solution of optimization problems, although multi-agent trajectories ensure high security, the computational complexity leads to an extension of path planning time. Figure 12e shows the formation maneuvering trajectory results based

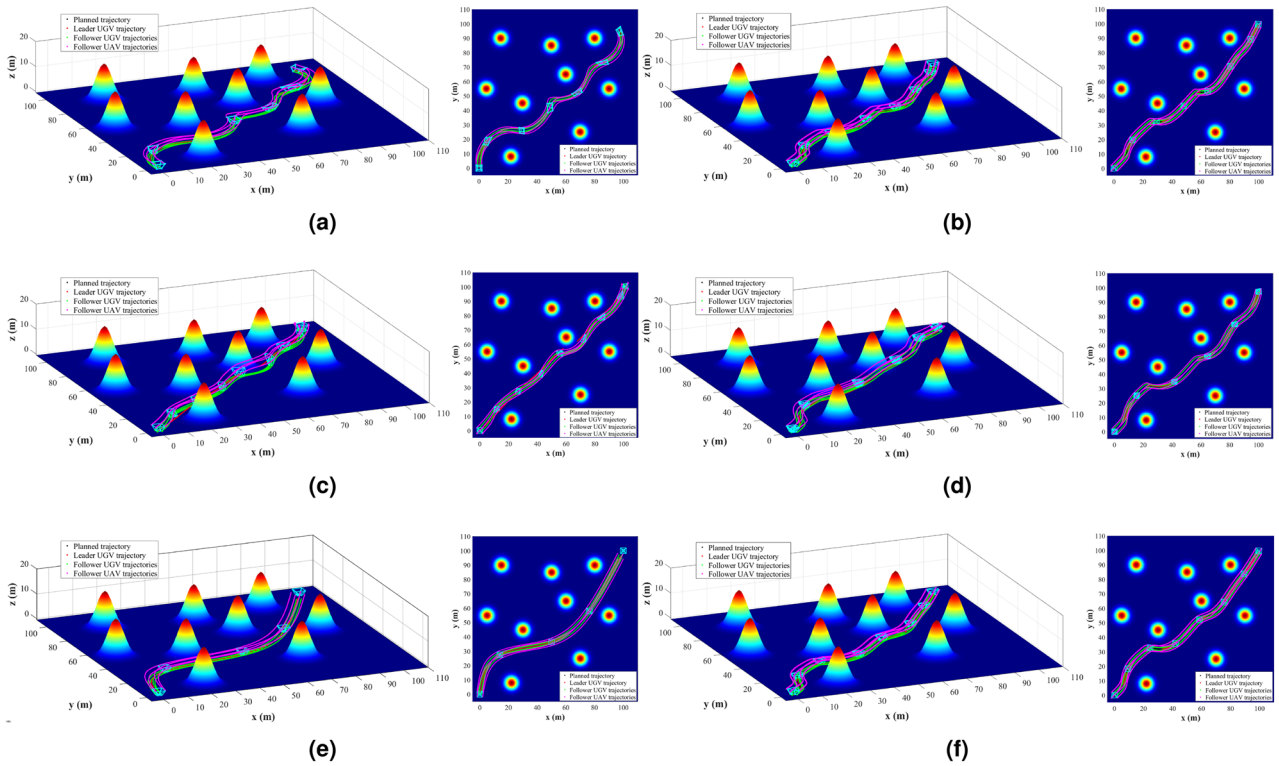


Fig. 12. **a** The distance errors $e_{ij}(t), i, j \in \mathscr{V}^*$ between any two agents. **b** The formation control input $\bar{\tau}_i$ for each agent in *x*, *y*, and *z* directions. **c** Sample of parameters estimates for leader UGV 9, $\hat{\phi}_9$. **d** Adaptive variation curve of repulsive force coefficient k_{rep}.

on the NMPC algorithm. Compared with other algorithms, although this algorithm shows better advantages in obstacle avoidance effect and formation maintenance, it is difficult to solve, takes a long time to calculate and has poor real-time performance. In contrast, Fig. 12f demonstrates the results of the proposed RG-ASAPF algorithm. This approach not only enables rapid planning of short and smooth trajectories but also ensures that the UAV–UGV formation maintains stable configuration throughout the process, thereby completing tasks efficiently with reduced energy consumption.

A comparison of quality indicators is provided in Table 5, which includes whether the formation can reach the goal point collision-free, computational time, planned trajectory length (*L1*), leading UGV trajectory length (*L2*), total trajectory length of all agents (*L3*), global tortuosity of planned path (*GS*), maximum tortuosity of planned path (*LS*), and total energy consumption of all agents. In terms of computational time, the traditional APF method requires 88.26 s, the improved APF algorithm requires 84.58 s, the PSO algorithm requires 18.62 s, the MPC algorithm requires 1224.73 s, the NMPC algorithm requires 1710.10 s, and the designed RG-ASAPF requires 75.08 s. In terms of planned trajectory length, the traditional APF is 162.67 m, the improved APF algorithm is 147.91 m, the PSO algorithm is 149.64 m, the MPC algorithm is 142.77 m, the NMPC algorithm is 158.34 m, while the designed RG-ASAPF is 144.62 m. For the leader UGV, the path lengths are 162.88 m (traditional APF), 147.95 m (improved APF), 149.90 m (PSO), 142.09 m (MPC), 158.95 m (NMPC), and 144.54 m (RG-ASAPF). For the total formation vehicles, the total path lengths are 1465.92 m (traditional APF), 1331.21 m (improved APF), 1350.28 m (PSO), 1282.30 m (MPC), 1431.47 m (NMPC), and 1301.38 m (RG-ASAPF). In terms of path smoothness, quantified by path GS and LS, the RG-ASAPF algorithm's smoothness significantly outperforms the traditional APF, the improved APF, PSO, and MPC based algorithm. Lower values indicate smoother trajectories with fewer redundant turns. In terms of energy consumption, the RG-ASAPF algorithm consumes 1938.22 J, representing a 10.6% reduction compared to the traditional APF algorithm, 0.8% reduction compared to the improved APF algorithm, 2.7% reduction compared to the PSO algorithm, 1.4% increase relative to MPC, and 7.0% reduction versus the NMPC-based algorithm. This highlights its overall energy-efficient performance. Overall, the proposed RG-ASAPF algorithm excels in critical metrics such as energy consumption, path length, planning speed, and trajectory smoothness. Its superior global optimization capability and formation stability offer an efficient, energy-saving solution for motion planning in heterogeneous UAV–UGV swarms.

In summary, based on the above simulation and analysis, the designed RG-ASAPF formation motion planning approach can effectively realize collision avoidance between the UGVs, between the UAVs, and between the UAV–UGV formation system and obstacles, achieve time-varying scaling reconfiguration of the formation, and can plan a collision-free and safe path for multi-UAV–UGV formation to arrive at the target point in dense obstacle environments. More importantly, the simulation results confirm that the designed RG-ASAPF method can complete formation navigation and motion planning tasks under different scenarios, particularly in dense obstacle environments, demonstrating the superiority and broad applicability of the algorithm. Moreover, simulation results confirm the method's robustness across diverse scenarios, particularly in coordinating multi-UAV–UGV formations amid dense obstacles. Furthermore, comparative analyses reveal that the RG-ASAPF algorithm outperforms the MPC algorithm, PSO-based algorithm, and traditional APF method in key metrics, including path planning efficiency, trajectory smoothness, formation stability, and energy consumption control. These results underscore the superiority and practical applicability of the proposed approach in UAV–UGV formation control and motion planning.

Conclusion and future work

This article develops a formation motion planning scheme for multiple heterogeneous UAVs and UGVs with undirected topologies subject to formation control and obstacle avoidance. By combining backstepping control theory, rigid graph theory, and ASAPF, the RG-ASAPF scheme is developed to address formation motion planning in dense obstacle environments, and the stability of the designed control scheme is rigorously guaranteed by Lyapunov stability theory. Finally, the simulation results indicate the feasibility and effectiveness of the proposed scheme as the multi-UAV–UGV cooperative system achieves motion planning and formation maintenance, successfully avoids obstacles, and reaches the target point while preserving the desired formation. In the current study, a designated leader is employed to plan the global maneuvering path of the formation. However, a failure of the leader may cause the formation to lose maneuverability or even result in obstacle

Algorithm	APF	IAPF	PSO	MPC	NMPC	RG-ASAPF
Goal reached	Success	Success	**Fail**	Success	Success	Success
Time t [s]	88.26	84.58	**18.62**	1224.73	1710.70	75.08
$L1$ [m]	162.67	147.91	149.64	**142.77**	158.34	144.62
$L2$ [m]	162.88	147.95	149.90	**142.09**	158.95	144.54
$L3$ [m]	1465.92	1331.21	1350.28	**1282.30**	1431.47	1301.38
GS [°]	0.066	0.047	0.038	0.055	**0.030**	0.032
LS [°]	5.15	0.27	1.75	0.27	**0.22**	0.25
E [J]	2167.97	1954.39	1992.48	**1911.82**	2084.08	1938.22

Table 5. Comparison of quality indicators obtained for the APF-, IAPF-, PSO-, MPC-, NMPC- and the proposed RG-ASAPF-based methods.



Fig. 13. One of the UGVs in the multi-UAV–UGV experimental platform.

collisions. To address this limitation, future research will focus on considering that while planning the overall maneuvering path for the formation, each following agent can plan its own path, and then integrate the leader' path and the followers' paths to design an overall maneuvering path for the formation. By fusing the leader and follower paths, a more robust and fault-tolerant formation maneuvering framework can be established. Meanwhile, extending the proposed method to soft landing problem of UAV on UGV in dynamic environments with randomly moving obstacles represents another important research direction. Furthermore, although the effectiveness of the proposed algorithm has been validated through simulations, but experimental verification remains limited due to hardware constraints. To overcome this issue, future research will focus on developing a dedicated experimental platform, as shows in Fig. 13, to systematically evaluate the algorithm's performance in real-world scenarios.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

Qian Chen: Conceptualization of this study, Methodology, Software, Writing-Original draft preparation. Xiufeng Liu: Data curation, Formal analysis, Investigation, Methodology, Writing-review and editing. Wendi Nie: Data curation, Investigation, Project administration, Supervision, Writing-review and editing. Haibing Liu: Data curation, Investigation, Resources, Visualization, Writing-original draft. Wen Pang: Data curation, Writing-Original draft preparation.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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